S5K3B6SA

Automotive 1/2.4" 3MP CMOS Image Sensor with On-Chip HDR & LFM

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MEKRA LANG

Data Sheet (Preliminary)

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Chip Handling Guide

Precaution against Electrostatic Discharge

When using semiconductor devices, ensure that the environment is protected against static electricity:

1. Wear antistatic clothes and use earth band.
2. All objects that are in direct contact with devices must be made up of materials that do not produce static electricity.
3. Ensure that the equipment and work table are earthed.
4. Use ionizer to remove electron charge.

Contamination

Do not use semiconductor products in an environment exposed to dust or dirt adhesion.

Temperature/Humidity

Semiconductor devices are sensitive to:

- Environment
- Temperature
- Humidity

High temperature or humidity deteriorates the characteristics of semiconductor devices. Therefore, do not store or use semiconductor devices in such conditions.

Mechanical Shock

Do not to apply excessive mechanical shock or force on semiconductor devices.

Chemical

Do not expose semiconductor devices to chemicals because exposure to chemicals leads to reactions that deteriorate the characteristics of the devices.

Light Protection

In non- Epoxy Molding Compound (EMC) package, do not expose semiconductor IC to bright light. Exposure to bright light causes malfunctioning of the devices. However, a few special products that utilize light or with security functions are exempted from this guide.

Radioactive, Cosmic and X-ray

Radioactive substances, cosmic ray, or X-ray may influence semiconductor devices. These substances or rays may cause a soft error during a device operation. Therefore, ensure to shield the semiconductor devices under environment that may be exposed to radioactive substances, cosmic ray, or X-ray.

EMS (Electromagnetic Susceptibility)

Strong electromagnetic wave or magnetic field may affect the characteristic of semiconductor devices during the operation under insufficient PCB circuit design for Electromagnetic Susceptibility (EMS).

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List of Conventions

Register RW Access Type Conventions

Type	Definition	Description
R	Read Only	The application has permission to read the register field. Writes to read-only fields have no effect.
W	Write Only	The application has permission to write in the register field.
RW	Read/Write	The application has permission to read and write in the register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0. The register field is updated immediately.
RW1	Read/Write	The application has permission to read and write in the register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0. The register field is updated after 1 frame.
RW2	Read/Write	The application has permission to read and write in the register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0. The register field is updated after 2 frames.

Register Value Conventions

Expression	Description
x	Undefined bit
X	Undefined multiple bits
?	Undefined, but depends on the device or pin status
Device dependent	The value depends on the device
Pin value	The value depends on the pin status

Reset Value Conventions

Expression	Description
0	Clears the register field
1	Sets the register field
x	Don't care condition

Warning: Some bits of control registers are driven by hardware or write operation only. As a result, the indicated reset value and the read value after reset should be different.

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List of Acronyms

Acronyms	Descriptions
ADLC	Auto Dark Level Compensation
APS	Active Pixel Sensor
CDS	Correlated Double Sampling
CIS	CMOS Image Sensor
FPN	Fixed Pattern Noise
WDR	Wide Dynamic Range (High Dynamic Range)
HSYNC	Horizontal Sync. Signal
OTP	One Time Programmable
PLL	Phase-Locked Loop
TG	Timing Generator
VSYNC	Vertical Sync. Signal

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1 Product Overview

This chapter includes the following sections:

- Introduction
- Features

1.1 Introduction

S5K3B6SA is a highly integrated 3MP camera chip that includes a CMOS Image Sensor (CIS) with image correction functionality and serial transmission capability using 4-lane MIPI. It is designed for delivering the full resolution pixels of 60 frames per second (fps) with 12-bit ADC, 2-exposure, intra-scene dual conversion gain, lateral overflow pixel and compressed outputs for High Dynamic Range (HDR) image with LED Flicker Mitigation.

S5K3B6SA is fabricated by the SAMSUNG 65 nm Back Deep Trench Isolation (BDTI) CMOS image sensor process, which is developed for imaging applications to realize a high-efficiency and low-power photo sensor. And it is also fabricated by the SAMSUNG 28nm logic process for low-power. The sensor consists of 1920 x 1536 active pixels and eight pixels on each side for color interpolation. This sensor meets 1/2.4" optical format with 3 MP resolution.

S5K3B6SA has on-chip 12-bit ADC arrays to digitize the pixel output and on-chip Correlated Double Sampling (CDS) to drastically reduce the Fixed Pattern Noise (FPN). It incorporates on-chip camera functions such as Bad Pixel Correction (BPC), exposure control, HDR reconstruction, and so on.

S5K3B6SA is programmable through an I2C serial interface and includes on-chip One-Time Programmable (OTP) Non-Volatile Memory (NVM).

S5K3B6SA is suitable for an automotive camera module with 2.8 V/1.8 V/1.05 V power supply, and the target applications are E-Mirror, Front/Rear-View camera and Surround-View-Monitoring (SVM) applications.

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Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

1.2 Features

S5K3B6SA supports the following features:

- 3 MP sensor with 1/2.4" optics
- Unit pixel size : 3.0 μm Split PD
- Effective resolution of 1920(H) x 1536 (V) with 3MP
- Active resolution of 1936 (H) x 1552 (V) with 3MP
  - Including eight pixels in each side for color processing
- High dynamic range and high sensitivity
- Electronic rolling shutter
- Maximum frame rate
  - Compressed 12-bit 3MP 60 fps (Intra-scene*1) Dual Conversion gain and Lateral overflow with LFM)
- Interfaces
  - Fine interface frequency control using additional dedicated PLL for EMI avoidance and integration flexibility
  - 2~4-lane selectable MIPI CSI-2 interface (Data rate : 4.0Gbps/lane)*2)
- Output formats
  - 24bit Combined RAW, 20/16/14/12-bit compressed combined RAW.
  - 20bit Combined RAW, 16/14/12-bit compressed combined RAW
- Control interface
  - CCI (Camera Control Interface) High speed I2C-compatible-Two-wire serial communication circuit up to 1MHz (FastMode+)
- 128 kbit (16 kbyte) on-chip OTP memories to support defect corrections and chip ID (38 kbit available for user)
- Analog gain up to x16
- Digital gain up to x256
- Vertical flip and horizontal mirror mode
- Static bad pixel correction with OTP
- Statistics data output for Combined image
- Output Defect and Detection map during image data stream. (RAW8)
- Built-in test pattern generation
- Embedded temperature sensor
- Embedded voltage monitoring
- Supply voltage of 2.8 V for pixel and analog, 1.8 V for I/O, and 1.05 V for digital core supply
- Satisfied ASIL B for compliance with funtional safety
- AEC-Q100 Grade2 Compatible(Tj = -40 ℃ to 125 ℃)

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* 1) Dual conversion gain readout within single frame.

* 2) Only 4 lanes are available to support maximum frame rate operation.

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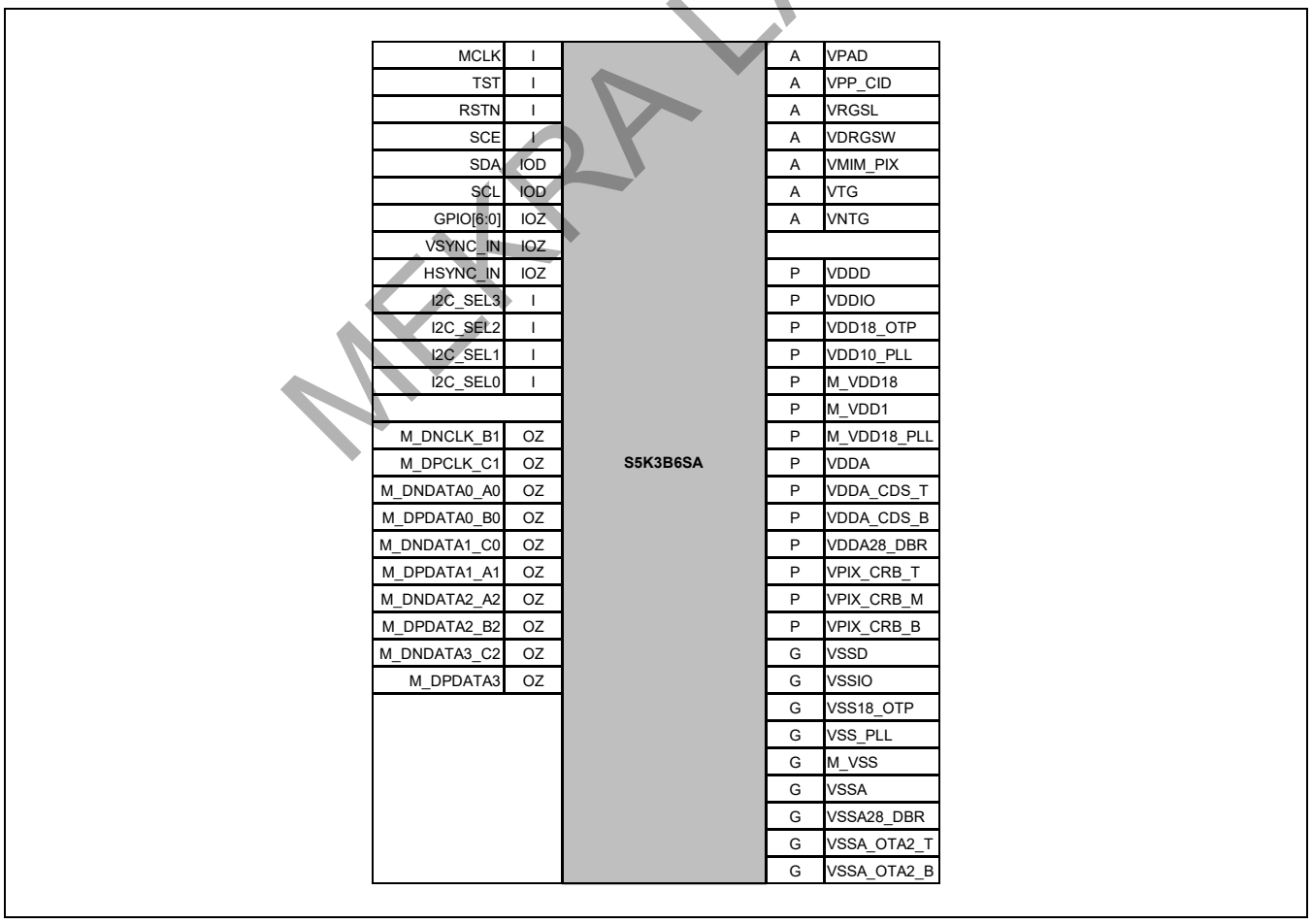
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2 Pin Configuration

This chapter includes the following sections:

- Simplified I/O Symbol Diagram
- Package Dimension
- Pin Configuration
- Pin Description
- Recommended Application circuit

2.1 Simplified I/O Symbol Diagram



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Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 2-1 Symbol Diagram of Simplified I/O

NOTE: Following are the legends used in *Figure 2-1*:

- I: Input
- O: Output
- OZ: Tri-state output
- IOZ: Tri-state in/out
- OD: Open-drain output
- IOD: Open-drain in/out
- P: Power and ground
- A: Analog test or external capacitor node

2.2 Package Dimension

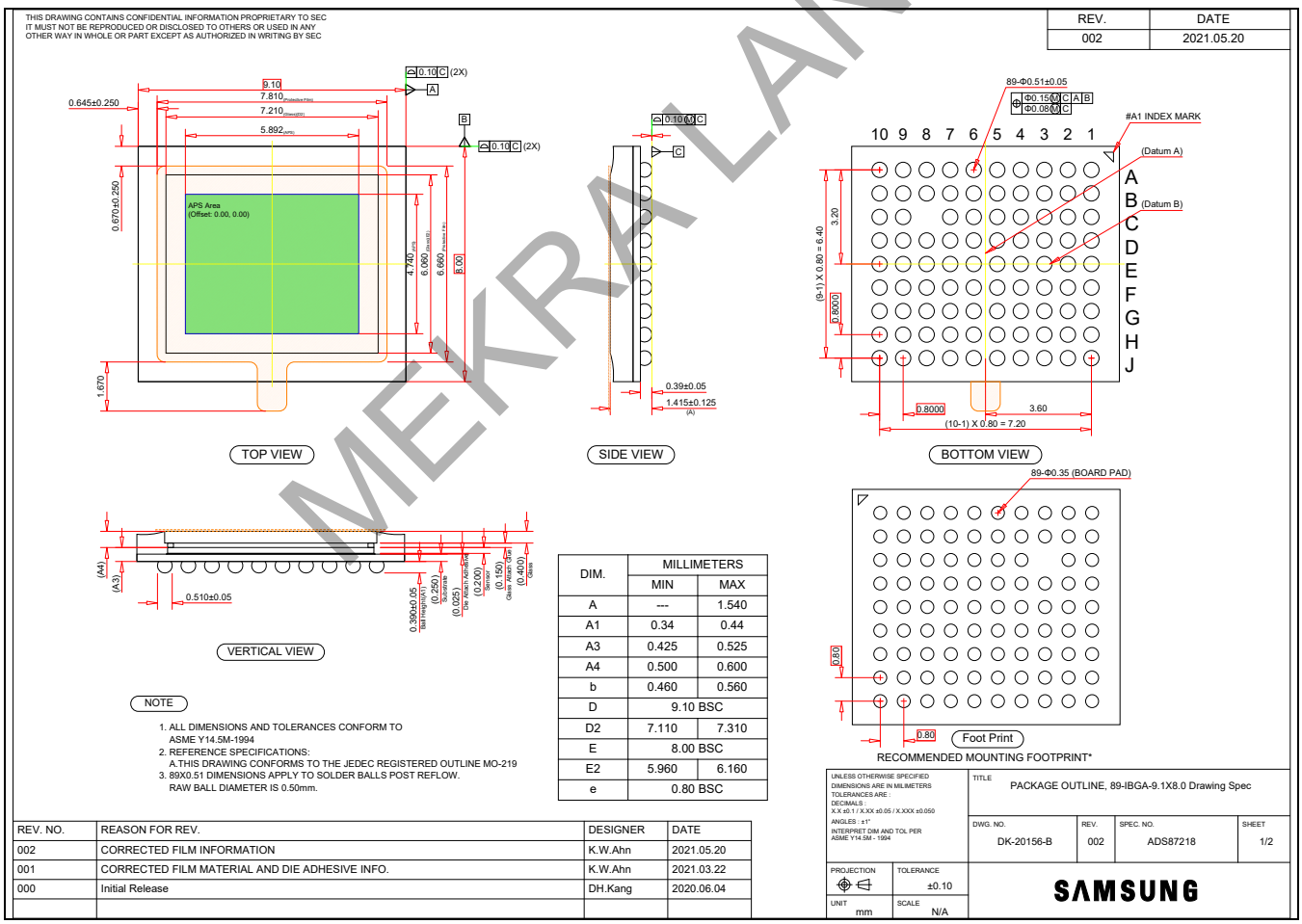


Figure 2-2 iBGA Package Dimension

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2.3 Pin Configuration

S5K3B6SA is assembled in iBGA package.

	1	2	3	4	5	6	7	8	9	10
A	VSSA28_ATOP	VSSA28_ATOP	VDDA28_ATOP	VPP_CID	SDA	RSTN	TST	VSSD	VDDA28_ATOP	VSSA28_ATOP
B	VSSD	VDDD	VDDA28_ATOP	VSSD	SCL	VSYNC_IN	HSYNC_IN	VSSD	VDDA28_ATOP	VSSA28_ATOP
C	M_DNDATA0	M_DPDATA0	VDD18_MIPI_PLL	VDDIO	VSSD	I2C_SEL0	I2C_SEL1		VDDD	VDDD
D	M_DNDATA1	M_DPDATA1	VDD10_MIPI	VDDD	VDDD	GPIO0	VSSD	VSSD	VSSD	VSSD
E	M_DNCLK	M_DPCLK	VSS_MIPI	VDDD	VSSD	GPIO1	GPIO2	VDDD	VPAD	VDDA28_ATOP
F	M_DNDATA2	M_DPDATA2	VDD10_MIPI	AVDD18_PLL	VSSD	GPIO3	GPIO4	VDDD	VDDD	VSSA28_ATOP
G	M_DNDATA3	M_DPDATA3	VSS_MIPI	MCLK	VSSD	GPIO5	GPIO6	I2C_SEL2	VSSD	VSSD
H	VSSD	VDDD	VSSIO	VDDIO	VDDD	VLTG	VRD	I2C_SEL3	VLNTG	VDDA28_ATOP
J	VSSA28_ATOP	VDDA28_ATOP	SCE	VSSD	VSNTG	VSTG	VRG	VSSA28_DBR	VDDA28_DBR	VSSA28_ATOP

Figure 2-3 Pin Configuration

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2.4 Pin Description

Table 2-1 Package Pin Description

Pin No	Pin Name	Pin Type	Description
A1	VSSA28_ATOP	GROUND	Analog ground
A2	VSSA28_ATOP	GROUND	Analog ground
A3	VDDA28_ATOP	POWER	Analog power - 2.8V
A4	VPP_CID	POWER	OTP power for PGM
A5	SDA	IN/OUTPUT	I2C data
A6	RSTN	INPUT	Reset input pad
A7	TST	INPUT	Test mode control - Tie low in normal operation
A8	VSSD	POWER	Digital core ground

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Pin No	Pin Name	Pin Type	Description
A9	VDDA28_ATOP	POWER	Analog power - 2.8V
A10	VSSA28_ATOP	GROUND	Analog ground
B1	VSSD	POWER	Digital core ground
B2	VDDD	POWER	Digital core power - 1.05V
B3	VDDA28_ATOP	POWER	Analog power - 2.8V
B4	VSSD	POWER	Digital core ground
B5	SCL	INPUT	I2C clock
B6	VSYNC_IN	INPUT	Frame (Vertical) sync. input
B7	HSYNC_IN	INPUT	Line (Horizontal) sync. input
B8	VSSD	POWER	Digital core ground
B9	VDDA28_ATOP	POWER	Analog power - 2.8V
B10	VSSA28_ATOP	GROUND	Analog ground
C1	M_DNDATA0	OUTPUT	MIPI data lane 0 negative
C2	M_DPDATA0	OUTPUT	MIPI data lane 0 positive
C3	VDD18_MIPI_PLL	POWER	MIPI analog power - 1.8V
C4	VDDIO	POWER	I/O power - 1.8 V (Default)
C5	VSSD	POWER	Digital core ground
C6	I2C_SEL0	INPUT	I2C ID selection pin 0
C7	I2C_SEL1	INPUT	I2C ID selection pin 1
C8	NC	-	No connection
C9	VDDD	POWER	Digital core power - 1.05V
C10	VDDD	POWER	Digital core power - 1.05V
D1	M_DNDATA1	OUTPUT	MIPI data lane 1 negative
D2	M_DPDATA1	OUTPUT	MIPI data lane 1 positive
D3	VDD10_MIPI	POWER	MIPI digital power - 1.05V
D4	VDDD	POWER	Digital core power - 1.05V
D5	VDDD	POWER	Digital core power - 1.05V
D6	GPIO0	OUTPUT	General purpose I/O
D7	VSSD	POWER	Digital core ground
D8	VSSD	POWER	Digital core ground
D9	VSSD	POWER	Digital core ground
D10	VSSD	POWER	Digital core ground
E1	M_DNCLK	OUTPUT	MIPI clock lane negative
E2	M_DPCLK	OUTPUT	MIPI clock lane positive
E3	VSS_MIPI	POWER	MIPI ground.

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Pin No	Pin Name	Pin Type	Description
E4	VDDD	POWER	Digital core power - 1.05V
E5	VSSD	POWER	Digital core ground
E6	GPIO1	OUTPUT	General purpose I/O
E7	GPIO2	OUTPUT	General purpose I/O
E8	VDDD	POWER	Digital core power - 1.05V
E9	VPAD	OUTPUT	Analog monitoring pad
E10	VDDA28_ATOP	POWER	Analog power - 2.8V
F1	M_DNDATA2	OUTPUT	MIPI data lane 2 negative
F2	M_DPDATA2	OUTPUT	MIPI data lane 2 positive
F3	VDD10_MIPI	POWER	MIPI digital power - 1.05V
F4	AVDD18_PLL	POWER	PLL analog power - 1.8V
F5	VSSD	POWER	Digital core ground
F6	GPIO3	OUTPUT	General purpose I/O
F7	GPIO4	OUTPUT	General purpose I/O
F8	VDDD	POWER	Digital core power - 1.05V
F9	VDDD	POWER	Digital core power - 1.05V
F10	VSSA28_ATOP	GROUND	Analog ground
G1	M_DNDATA3	OUTPUT	MIPI data lane 3 negative
G2	M_DPDATA3	OUTPUT	MIPI data lane 3 positive
G3	VSS_MIPI	POWER	MIPI ground.
G4	MCLK	INPUT	External clock input
G5	VSSD	POWER	Digital core ground
G6	GPIO5	OUTPUT	General purpose I/O
G7	GPIO6	OUTPUT	General purpose I/O
G8	I2C_SEL2	INPUT	I2C ID selection pin 2
G9	VSSD	POWER	Digital core ground
G10	VSSD	POWER	Digital core ground
H1	VSSD	POWER	Digital core ground
H2	VDDD	POWER	Digital core power - 1.05V
H3	VSSIO	GROUND	I/O ground
H4	VDDIO	POWER	I/O power - 1.8V (Default)
H5	VDDD	POWER	Digital core power - 1.05V
H6	VLTG	POWER	High voltage level for pixel transfer gate
H7	VRD	POWER	MIM anode node for small PD
H8	I2C_SEL3	INPUT	I2C ID selection pin 3

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Pin No	Pin Name	Pin Type	Description
H9	VLNTG	POWER	Negative voltage level for pixel transfer gate
H10	VDDA28_ATOP	POWER	Analog power - 2.8V
J1	VSSA28_ATOP	POWER	Analog ground
J2	VDDA28_ATOP	POWER	Analog power - 2.8V
J3	SCE	INPUT	Clock selection (external/internal)
J4	VSSD	POWER	Digital core ground
J5	VSNTG	POWER	Negative voltage level for Small pixel transfer gate
J6	VSTG	POWER	Positive pixel control voltages of RG for DCG and SW for small PD.
J7	VRG	POWER	High voltage level for pixel reset/selection transistors
J8	VSSA28_DBR	POWER	Doubler ground
J9	VDDA28_DBR	POWER	Analog power for voltage doubler - 2.8V
J10	VSSA28_ATOP	POWER	Analog ground

2.5 Recommended Application Circuit

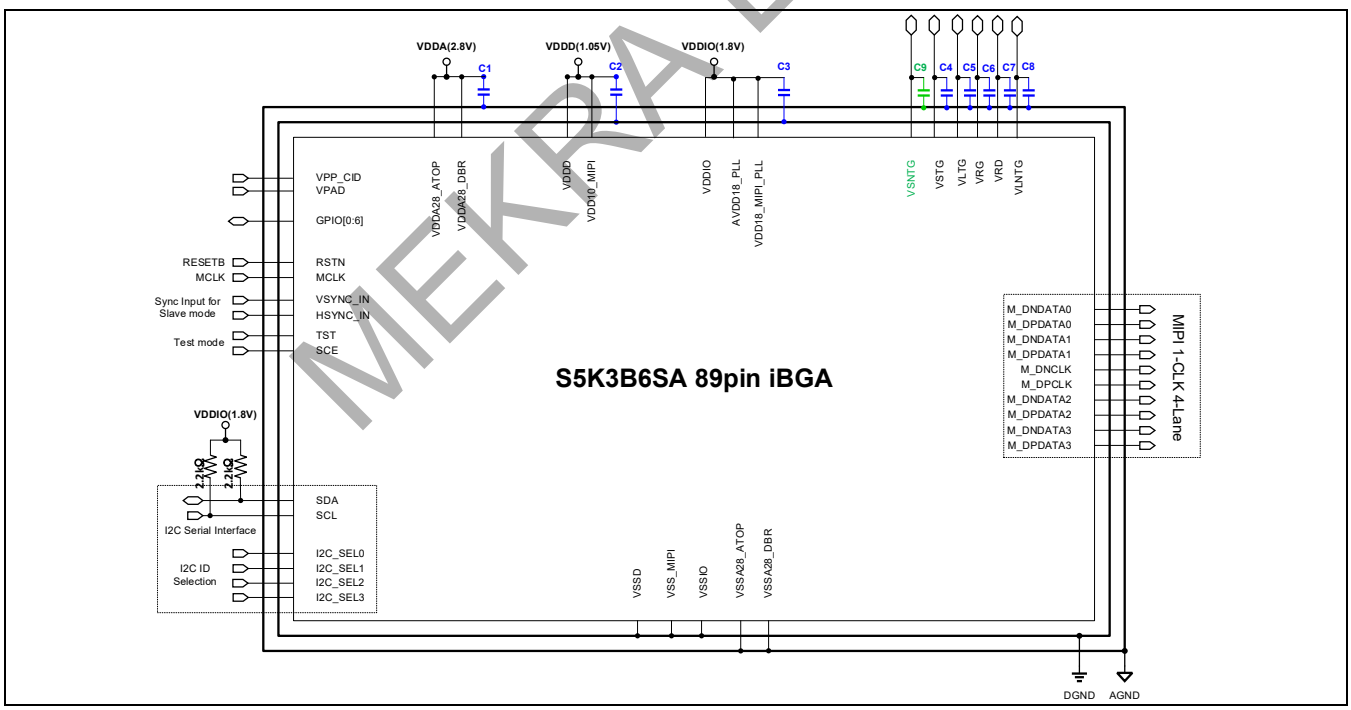


Figure 2-4 iBGA Application Circuit of MIPI

Table 2-2 Recommend Capacitor to Design a Module PCB

Capacitor No.	Power	Capacitor(μF)

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C1	VDDA(VDDA28_ATOP, VDDA28_DBR)	4.7 μF
C2	VDDD(VDDD, VDD10_MIPI)	2.2 μF
C3	VDDIO(VDDIO, AVDD18_PLL, VDD18_MIPI_PLL)	2.2 μF
C4	VSTG	0.1 μF
C5	VLTG	0.1 μF
C6	VRG	4.7 μF
C7	VRD	10.0 μF
C8	VLNTG	4.7 μF
C9*	VSNTG	4.7 μF

NOTE: 1. The digital ground (DGND) and the analog ground (AGND) can be merged unless both ground resistances are large for long flexible PCB.
2. RESETB(RSTN) indicates reset active low.
3. TST and SCE input must be tied to digital ground (DGND).
4. VSYNC_IN / HSYNC_IN input must be tied to digital ground (DGND) if it is not used.
5. I2C_SEL0 / I2C_SEL1 / I2C_SEL2 / I2C_SEL3 input must be tied to ground (DGND) or power (VDDIO)
6. All external capacitors must be connected as close as possible to each PAD of chip side.
7. The differential characteristic impedance of the MIPI transmission line pair is 100Ω.

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3 Power sequence

This chapter includes the following sections:

- System state description and diagram
- Power-up/down sequence
- Standby/wake-up sequence
- Sensor system reset

3.1 System State

The Sensor module has five operating modes, such as power-off, hardware standby, software standby, streaming and safety state as described in *Table 3-1*. Transition from one state to another is achieved by issuing the appropriate state command through the CCI serial control interface, the RSTN (XSHUTDOWN) signal changing state, and the power supplies. By default, S5K3B6SA is powered up by enabling the CSI-2 serial data interface. *Figure 3-1* illustrates the valid state changes for the sensor module.

S5K3B6SA supports the safety state to ensure the prohibition data stream when there is an error occurred from the error detection logic. Sensor transitions its state from safety state to software standby if an error is cleared through registers. Then, sensor can stream data.

If an error occurs during the data streaming, S5K3B6SA outputs the last data normally and then transitions to safety state.

Table 3-1 Operating Mode Summary

System State	Description
Power-off	Power supplies are turned off.
Hardware standby	RSTN is low. Power supplies are turned on, but cannot communicate with the sensor. Internal registers are set to default values.
Software standby	RSTN is high. Register setting through I2C is possible.
Streaming	The sensor is fully functional and video outputs are available.
Safety state	Sensor enters into safety state when there is a fault in sensor operation. • It cannot stream data in this state until the error flag is reset. Sensor transitions to software standby state after the error flag is reset.

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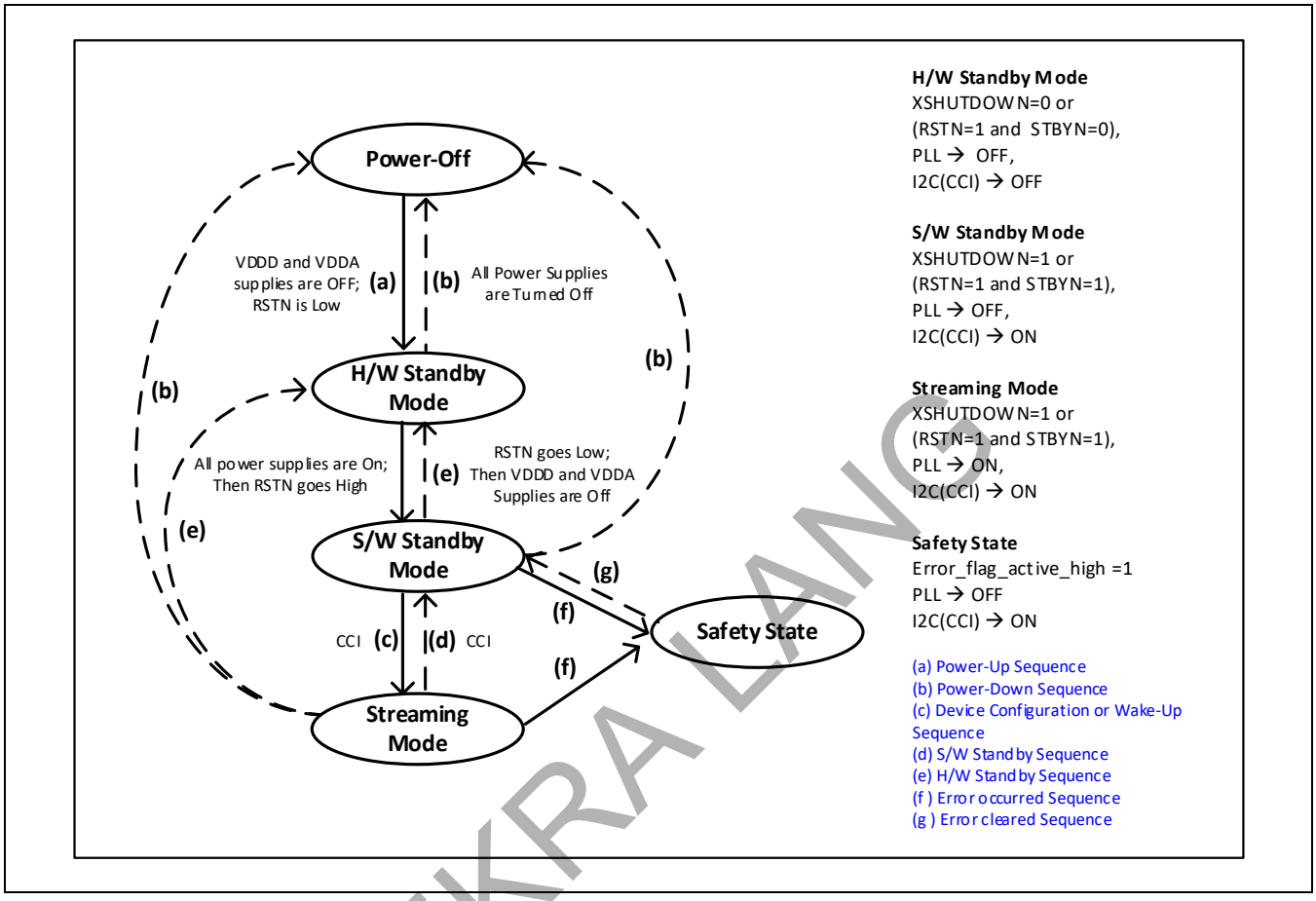


Figure 3-1 Operating Mode Diagram

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3.2 Power-Up Sequence

The digital and analog supply voltages can be powered up in any order, such as first VDDD and then VDDA or first VDDA and then VDDD.

After the digital and analog voltage supplies are powered up, RSTN (XSHUTDOWN) should be low, and the sensor module transitions to hardware standby mode. Until RSTN is low and VDDD is down, the sensor module stays in hardware standby mode.

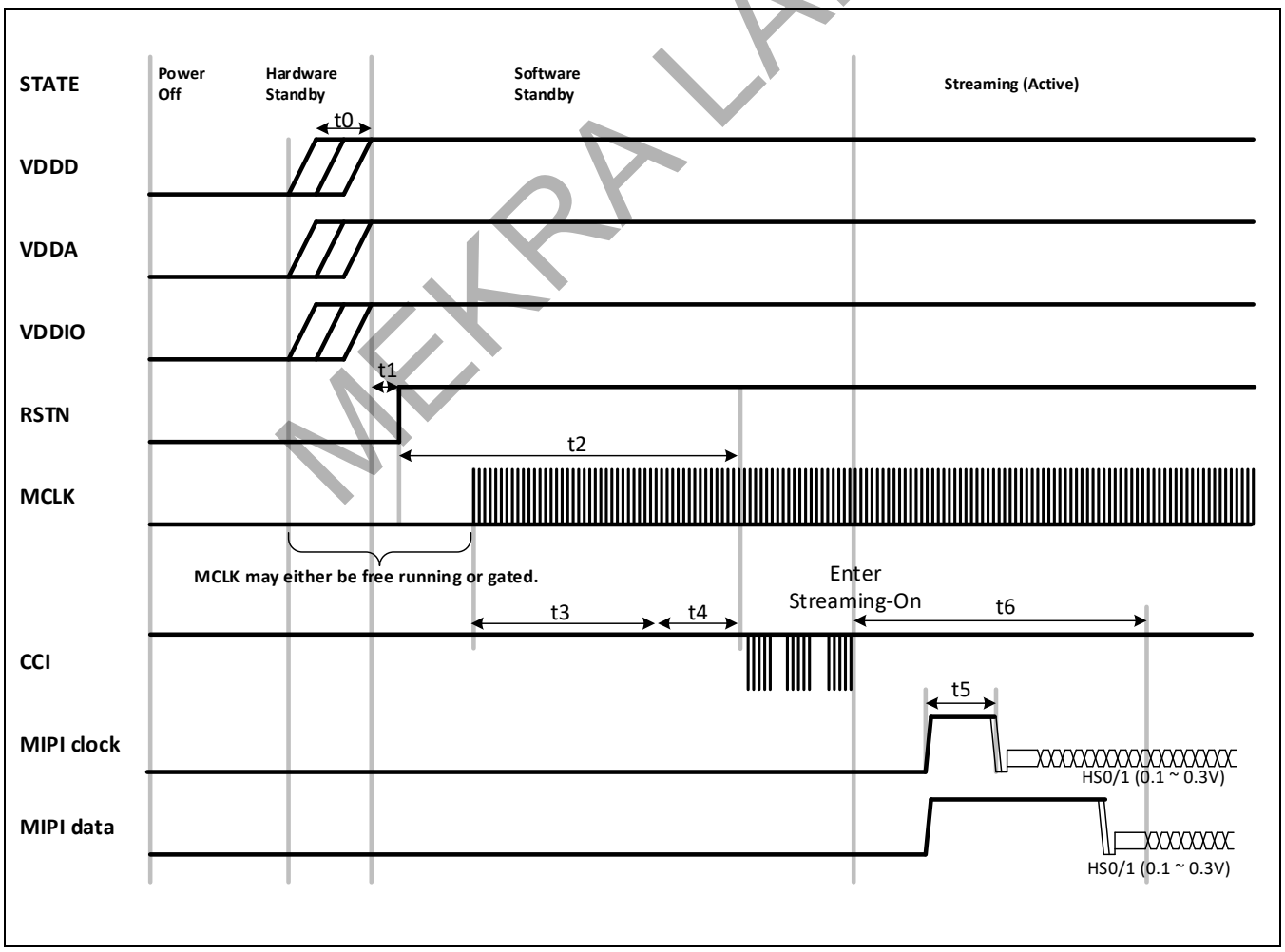
The assertion of RSTN ensures that the CCI register values are correctly initialized to their default values.

When RSTN goes high, all PADs exit from fail-safe mode and transitions to normal operating mode.

The MCLK can be free running or gated before it is enabled during software standby mode.

NOTE:

1. In hardware standby mode, external VDDD and VDDA should be turned off.
2. To reduce the power consumption, it is recommended to turn off the external clock (MCLK) in software standby state.



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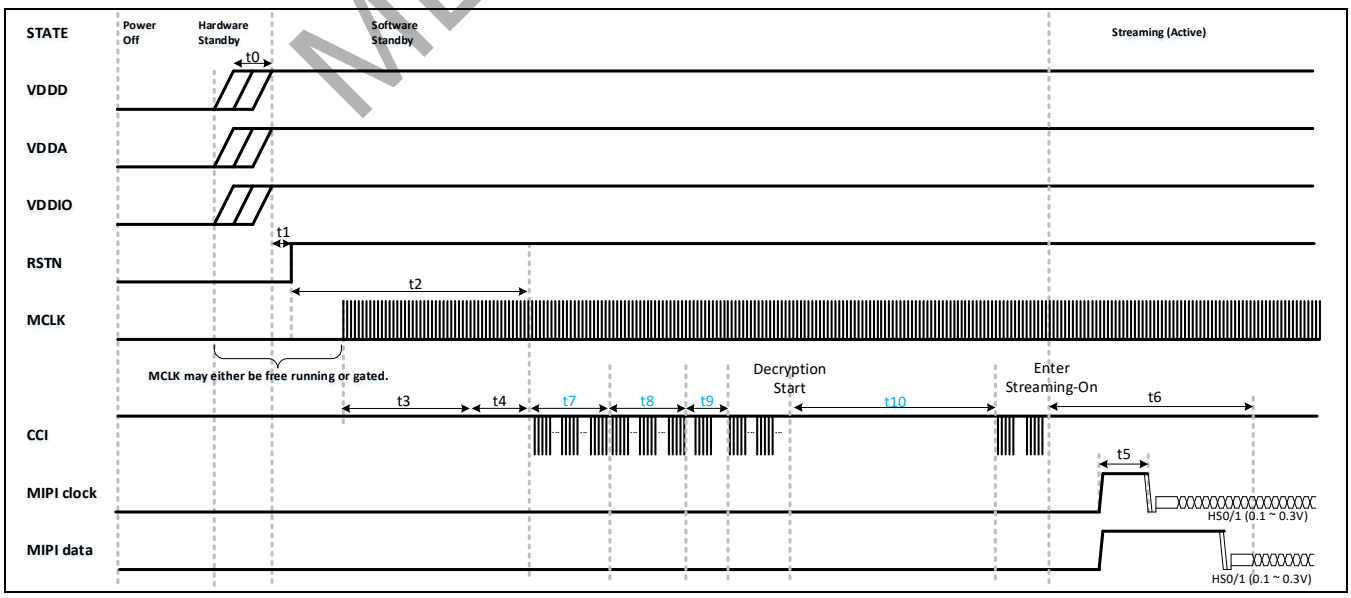
Figure 3-2 Power-Up Sequence Without Security IP

Table 3-2 Timing Constraints of Power-Up Sequence

Constant	Label	Min.	Max.	Unit
VDDA//VDDD/VDDIO rising time	t0	VDDA/VDDD/VDDIO rises in any order. The rising separation can vary from 0 ns to indefinite time.		ns
VDDD rising to RSTN (XSHUTDOWN) rising	t1	0.0	–	ns
RSTN (XSHUTDOWN) rising to first CCI transaction	t2	0.95	27.34	ms
RSTN rising to POST operation* *POST - Power On Self-Test (External clock 27 MHz)	t3	0.63 (Disable LBIST,MBIST)	17.66	ms
POST done to OTP operation	t4	0.32	9.68	ms
D-PHY initialization period (TINIT)	t5	0.1	–	ms
Entering to streaming mode - First frame of start sequence	t6	Internal readout delay + coarse integration time value		–

NOTE:

1. Timing Costraints for General Operation(t1~t6) are same as Power-Up Sequence without Security IP.
2. t7~t10 are required additionally for security function.
3. Image streaming can start before t7, since the additional sequence of RSA decryption(Session Key sharing) can be done also after image streaming. In this case, data authentication, which is only available after session key sharing, starts from the next frame after RSA decryption is finished.
4. Table 5 I2C timing values are estimated assuming to use I2C Fast Mode.



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Figure 3-3 Power-Up Sequence with Security IP

Table 3-3 Timing Constraints of Power-Up Sequence with Security Function Enabled

Period	Operation	Time	Description
t7	OTP read by ECU via I2C	RSA 3k: 59.5ms RSA 2k: 40ms	PubK, Certdevice, UID read by ECU to verify RSA Signature
t8	ESsk write by ECU via I2C	RSA 3k: 23.75ms RSA 2k: 16ms	I2C-write of ESsk(ciphertext) by ECU
t9	OTP re-load to RSA memory	RSA 3k: 938us RSA 2k: 625us	PubK & PrivK-load to RSA memory
t10	RSA decryption	RSA 3k: 233.9㎳(128.57㎒) / 1.113s(27㎒;MCLK) RSA 2k: 71.5㎳(128.57㎒) / 340㎳(27㎒;MCLK)	RSA Decryption time including RSA memory-read

NOTE: Time changed according to the clock speed.

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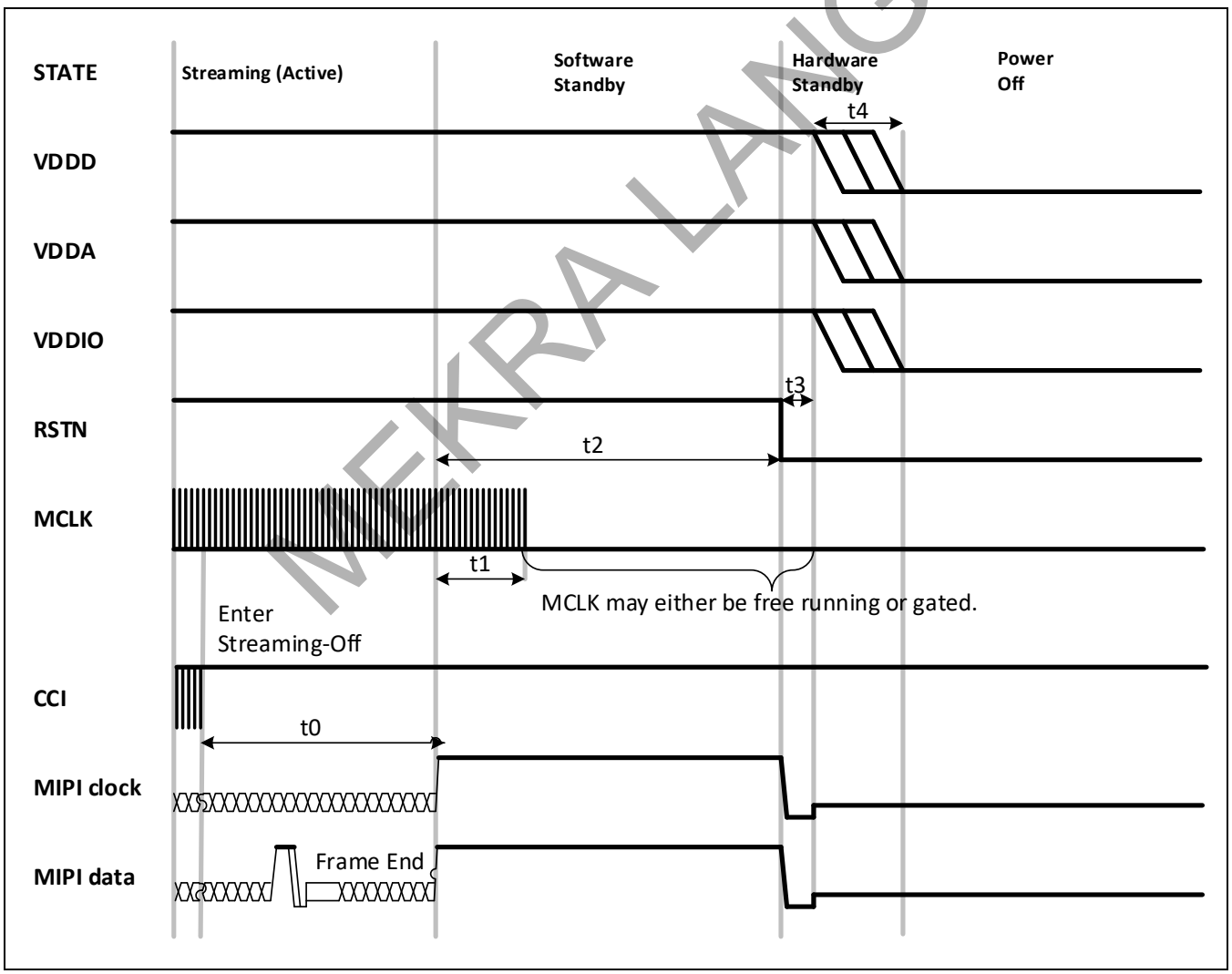
3.3 Power-Down Sequence

The digital and analog supply voltages can be powered down in any order, e.g., VDDD then VDDA or VDDA then VDDD.

Similar to the power-up sequence, the MCLK (EXTCLK): input clock may be either gated or continuous.

If the CCI command to exit streaming is received while a frame of valid active data is being output, then the sensor module must wait for the frame end code before entering software standby mode. Frame end code may come either after all frame pixels were transmitted or during the frame when next line transmission is completed - based on a configuration register (Refer to S5K3B6SA application notes for details)

If the CCI command to exit streaming mode is received during the inter frame time, then the sensor module must enter software standby mode immediately.



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Figure 3-4 Power-Down Sequence

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 3-4 Power-Down Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
Enter Software Standby CCI command- Device in Software Standby mode	t0	If outputting a frame of MIPI, the data waits for the MIPI frame end code before entering software standby; otherwise enter software standby mode immediately.		–
Minimum number of MCLK (EXTCLK) cycles after the last CCI transaction or MIPI frame end code.	t1	512	–	MCLK cycles
Last CCI Transaction or MIPI frame end code - RSTN (XSHUTDOWN) falling	t2	512	–	
RSTN (XSHUTDOWN) falling to VDDD or VDDA falling	t3	0.0	–	ns
VDDA/VDDA/VDDD/VDDIO falling time	t4	VDDA/VDDD/VDDIO may fall in any order. The rising separation can vary from 0 ns to indefinite		

NOTE: For minimal power at hardware standby mode external VDDD and VDDA should be OFF.

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4 Control Interface

This chapter includes the following sections:

- I2C
- CCI

4.1 I2C

S5K3B6SA supports the I2C communication method, which is the 2-wired serial interface between a backend processor (host) and a sensor, to control the register values.

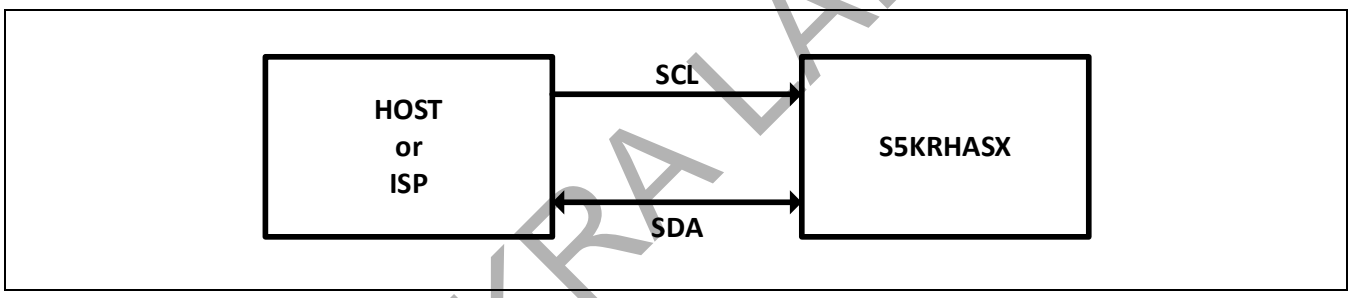


Figure 4-1 I2C Control Interface

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Table 4-1 describes the I2C signals.

Table 4-1 Pin Description of I2C

Pin Name	Description
SCL	Serial communication clock
SDA	Serial communication data

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4.2 CCI

S5K3B6SA supports the Camera Control Interface (CCI), which is an I2C Fast Mode+ compatible interface, to control the transmitter. S5K3B6SA always acts as a slave in CCI bus. CCI is capable of handling several slaves in the bus, but multi-master mode is not supported. Typically, only the receiver and transmitter are connected to CCI bus. This ensures a proper S/W implementation.

Typically, CCI is separated from system I2C bus, but I2C-compatibility ensures that it can connect the transmitter to system I2C bus. CCI is a subset of the I2C protocol including the minimum combination of obligatory features for the I2C slave device specified in the I2C specification. Therefore, transmitters complying with CCI specification are connected to system I2C bus. However, it is important to ensure that the I2C masters do not try to utilize these I2C features, which are not supported in transmitters that are complying with CCI specification. Each transmitter conformed to CCI specification has additional features to support I2C.

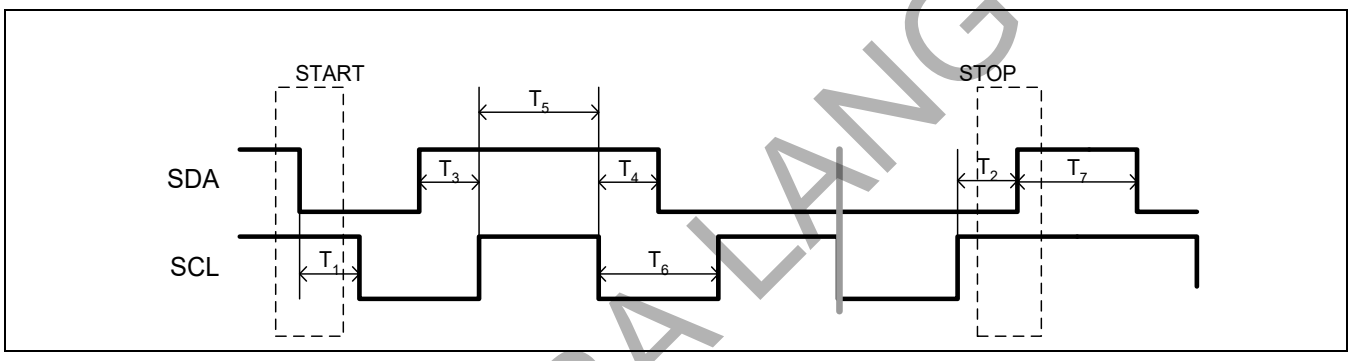


Figure 4-2 Timing Diagram of CCI

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Table 4-2 I2C Standard Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	100	kHz
Hold time for start condition	T_1	4.0	-	µs
Setup time for stop condition	T_2	4.0	-	
Data setup time	T_3	250	-	ns
Data hold time	T_4	0	3.45	µs
High period of the SCL clock	T_5	4.0	-	
Low period of the SCL clock	T_6	4.7	-	
Bus free time between stop and start conditions	T_7	4.7	-	
Rise time for both SDA and SCL signals	-	-	1000	ns
Fall time for both SDA and SCL signals	-	-	300	
Capacitive load for each bus line	CB	-	400	pF

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Table 4-3 I2C Fast Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	400	kHz
Hold time for start condition	T_1	0.6	-	µs
Setup time for stop condition	T_2	0.6	-	
Data setup time with external clock (MCLK) above 15 MHz	T_3	0.6	-	µs
Data hold time	T_4	0	0.9	µs
High period of the SCL clock	T_5	0.6	-	
Low period of the SCL clock	T_6	1.3	-	
Bus free time between stop and start conditions	T_7	1.3	-	
Rise time for both SDA and SCL signals	-	-	300	ns
Fall time for both SDA and SCL signals	-	-	300	
Capacitive load for each bus line	CB	-	400	pF

Table 4-4 I2C Fast Mode+ Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	1000	kHz
Hold time for start condition	T_1	0.26	-	µs
Setup time for stop condition	T_2	0.26	-	
Data setup time, external clock (MCLK) above 15 MHz	T_3	50	-	ns
Data hold time	T_4	0	-	µs
High period of the SCL clock	T_5	0.26	-	
Low period of the SCL clock	T_6	0.5	-	
Bus free time between stop and start conditions	T_7	0.5	-	
Rise time for both SDA and SCL signals	-	-	120	ns
Fall time for both SDA and SCL signals	-	-	120	
Capacitive load for each bus line	CB	-	550	pF

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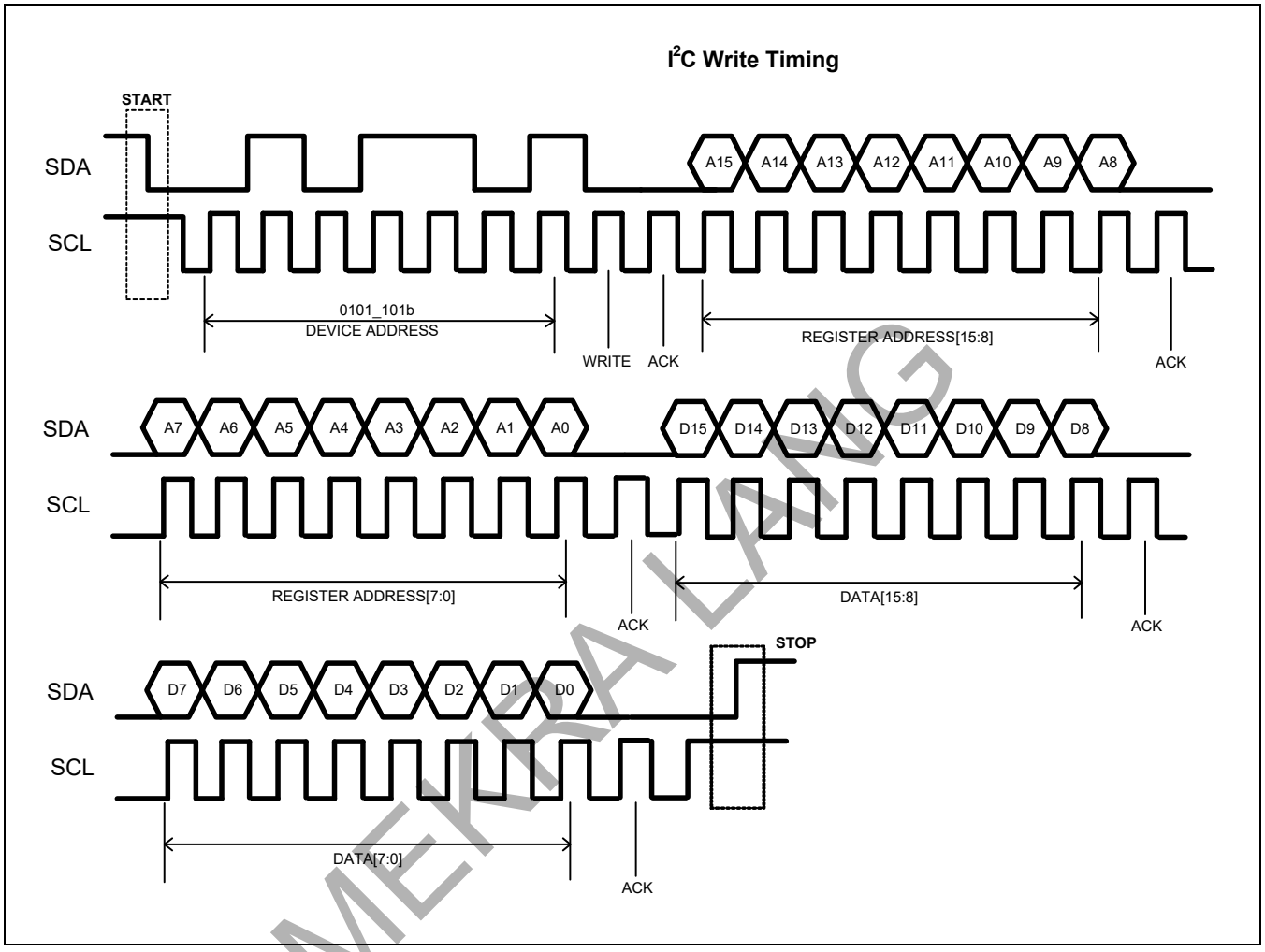
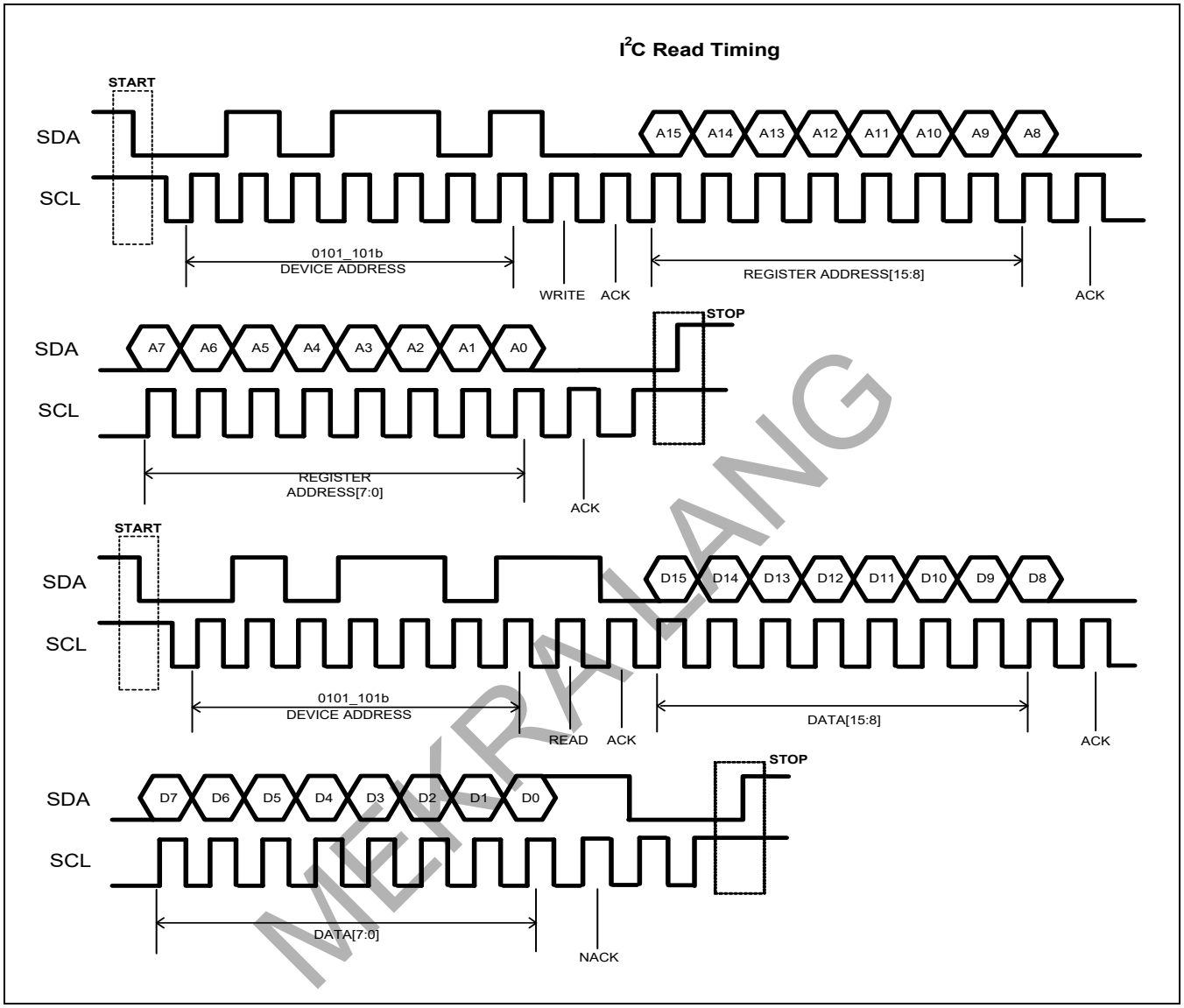


Figure 4-3 CCI Write Timing Example

NOTE: The device address can be changed by pin configuration of I2C_SEL, as described in the pad description.

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Figure 4-4 CCI Read Timing Example

NOTE: The device address can be changed by the pin configuration of I2C_SEL, as described in the pad description.

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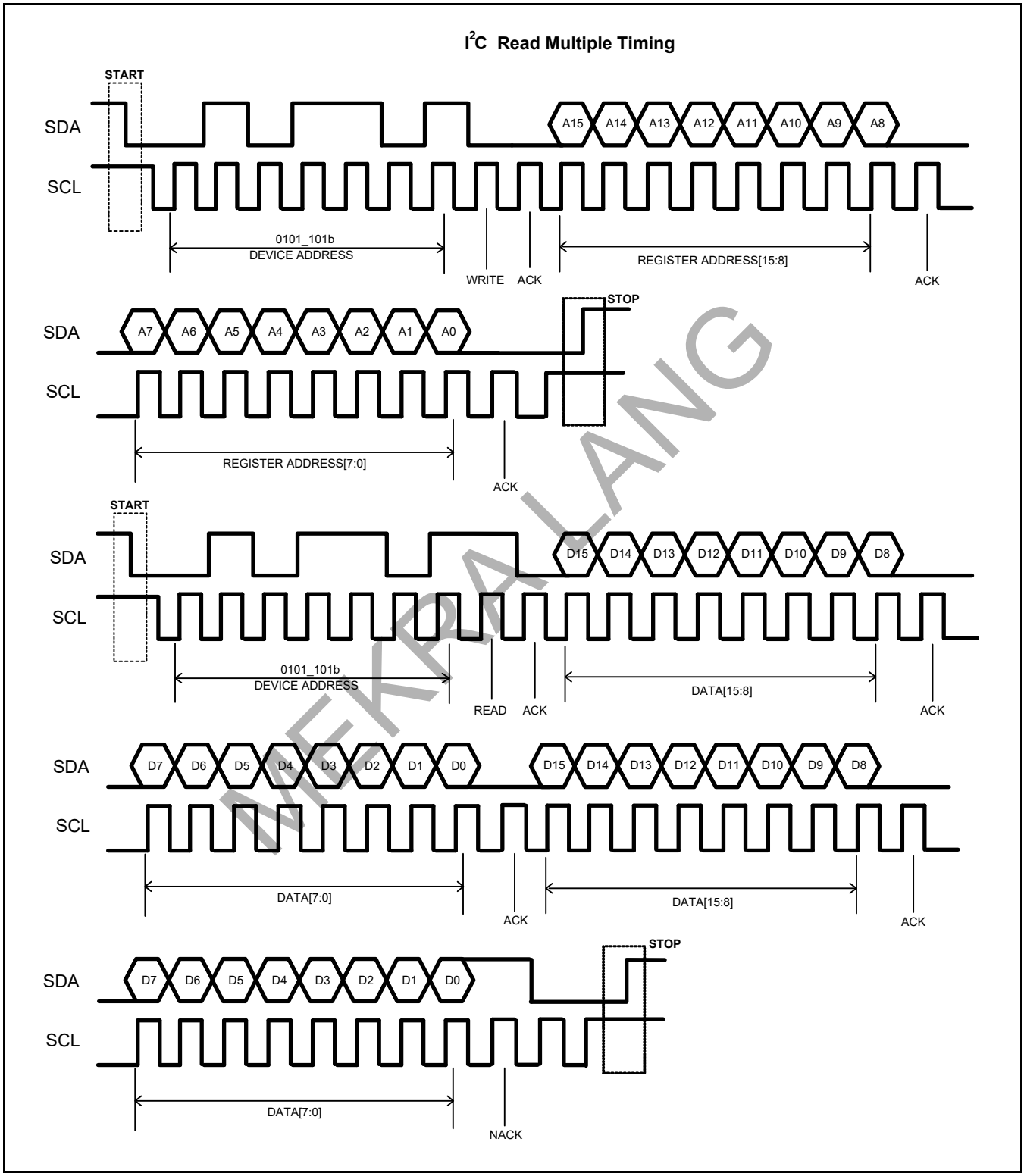


Figure 4-5 CCI Read Multiple Timing Example

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NOTE: The device address can be changed by the pin configuration of I2C_SEL, as described in the pad description.

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It is possible to configure up to eight I2C slave addresses using I2C_SEL3 ~ I2C_SEL0 pins.

Table 4-5 I2C ID Address (I2C_SEL3 ~ I2C_SEL0 Pad)

I2C_SEL[3:0]	Slave Address (7-bit + Read Mode)	Slave Address (7-bit + Write Mode)	Comment
0	0010_0001b/21h	0010_0000b/20h	Address 1
1	0011_0001b/31h	0011_0000b/30h	Address 2
2	0010_1001b/29h	0010_1000b/28h	Address 3
3	0011_1001b/39h	0011_1000b/38h	Address 4
4	0110_0001b/61h	0110_0000b/60h	Address 5
5	1010_0001b/A1h	1010_0000b/A0h	Address 6
6	1100_1001b/C9h	1100_1000b/C8h	Address 7
7	1110_1011b/EBh	1110_1010b/EAh	Address 8

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5 Pixel Information

This chapter includes the following sections:

- Pixel array
- Spectral sensitivity characteristics

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5.1 Pixel Array

The effective pixel array is 1920 (H) × 1536 (V). Although the total pixel array includes different kinds of pixels, pixel array can be read from (8, 48) to (1927, 1583). The pixels are covered by clear and red color filters arranged in RGB Bayer structure. Micro-lens are placed on the top of color filters to increase the effective fill factor for higher sensitivity.

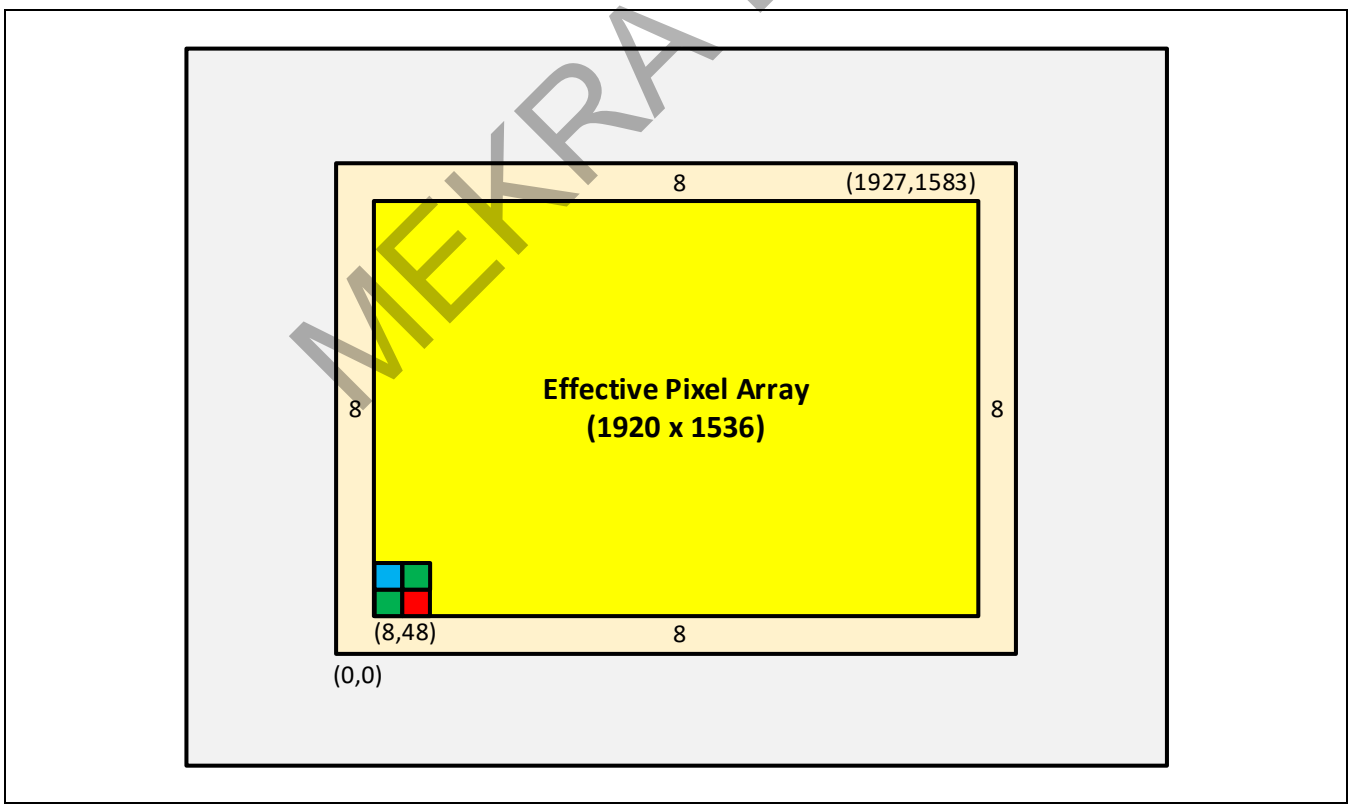


Figure 5-1 Pixel Array Information

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

NOTE: For top view of the chip, vertically flip the displayed image.

5.2 Spectral Sensitivity Characteristics

The spectral sensitivity of S5K3B6SA is characterized at the half signal output of the clear pixel on wavelength sweep. *Figure 5-2* shows the normalized sensitivity ratio of LPD and SPD.

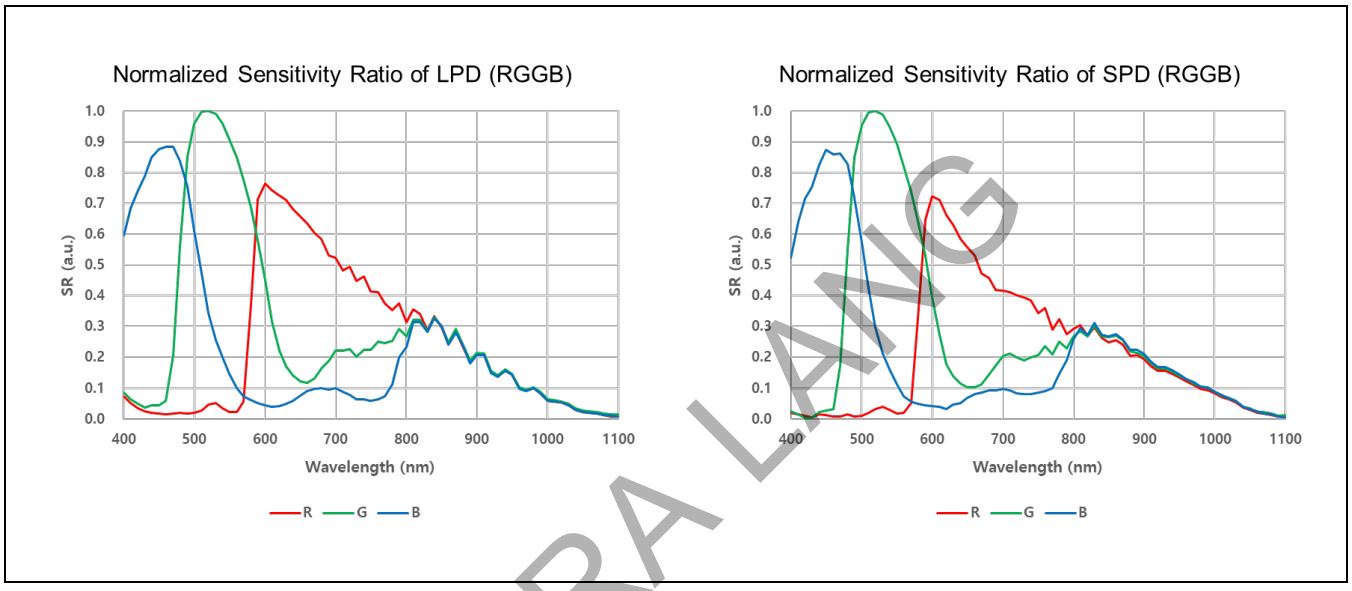


Figure 5-2 Spectral Sensitivity Characteristics

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NOTE: The spectral sensitivity can be changed to improve the performance without any notice before the mass production.

5.3 Quantum Efficiency Characteristics

Figure 5-3 show the quantum efficiency characteristics of RGGB, respectively. QE is the average value by using three samples.

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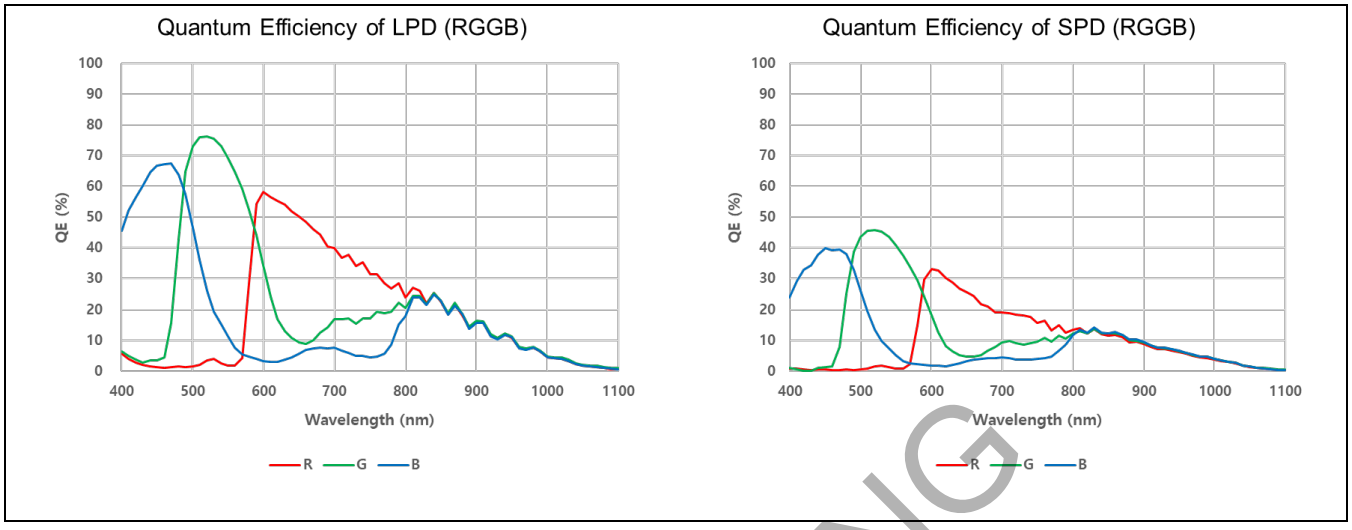


Figure 5-3 Quantum Efficiency of LPD and SPD of RGGB CF Type

5.4 Lens Specification (CRA: Chip Ray Angle Specification)

The target CRA due to the image height is described in *Figure 5-4* and *Table 5-1*.

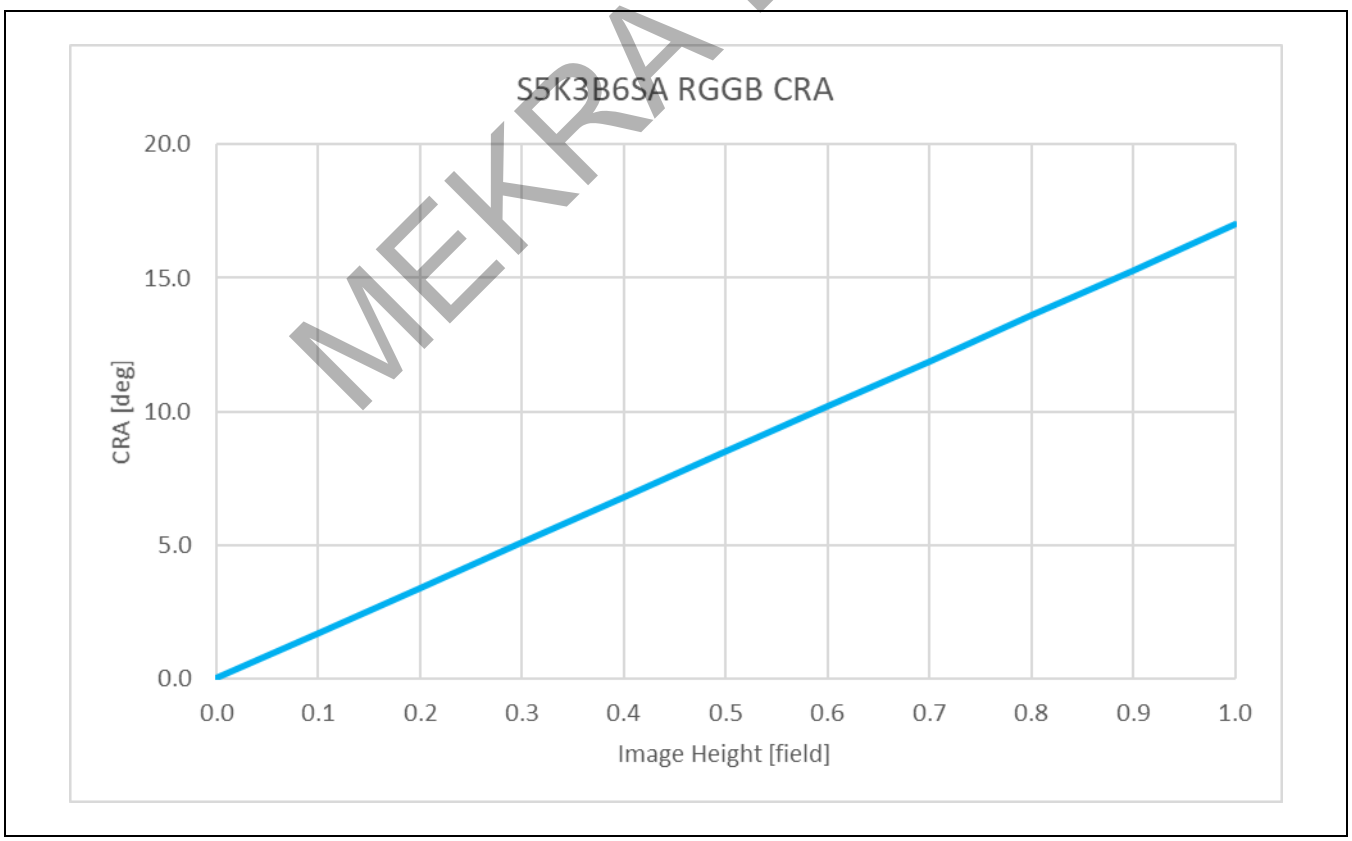


Figure 5-4 Target CRA

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Table 5-1 Target CRA

Image Heigt [field]	Image Heigt [mm]	RGGB CRA [deg]
0.0	0.00	0.0
0.1	0.37	1.7
0.2	0.74	3.4
0.3	1.11	5.1
0.4	1.48	6.8
0.5	1.84	8.5
0.6	2.21	10.2
0.7	2.58	11.9
0.8	2.95	13.6
0.9	3.32	15.3
1.0	3.69	17.0

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6 Functional Description

This chapter includes the following sections:

- Functional Block Diagraim
- PLL and clock generator
- Operation mode
- Slave mode
- Frame readout format
- Image orientation (H/V)
- Frame rate control
- Integration time control (Electronic shutter control)
- Gain control
- Auto dark level correction
- Image output format

S5K3B6SA provides several readout modes and specific functions for video systems and for the required variety of clocks. Two Phase-Locked Loops (PLLs) are embedded in the sensor. One generates clocks for systems, and the other one generates clocks for high-speed serial interfaces. The master input clock range is between 6 MHz and 64 MHz.

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6.1 Functional Block Diagram

S5K3B6SA is composed of an APS array, 12-bit ADCs, an embedded Timing Generator (TG), pixel data process blocks, OTP controller, two Phase-Locked Loop of PLL1 and PLL2, Automotive Signal Processing block which supports interleaving, merge and compress for HDR data and interface blocks.

Figure 6-1 illustrates the functional block diagram of sensor.

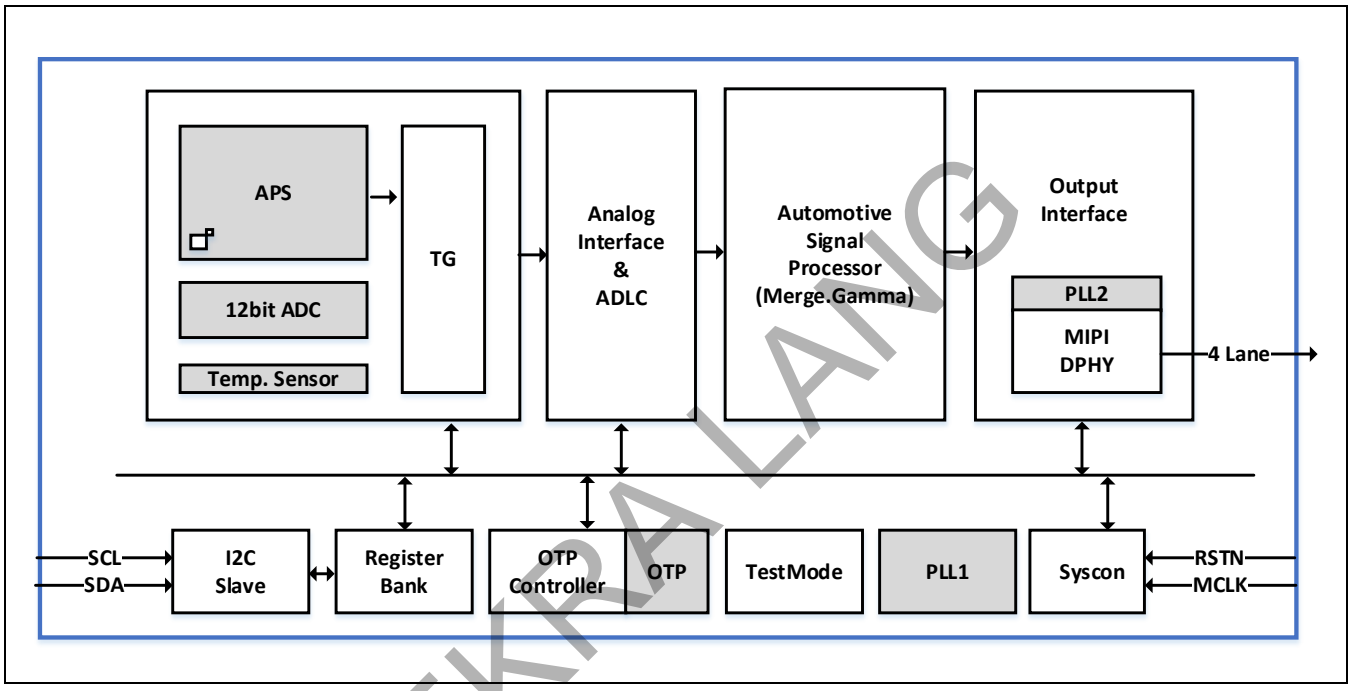


Figure 6-1 Functional Block Diagram of Sensor

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6.2 PLL and Clock Generator

S5K3B6SA contains two PLLs and a clock generator, which generate all the necessary video timing, output pixel clocks, and interface serializing clocks.

By setting the divide-ratio for pre PLL clock divider (`pll_sys_p and pll_int_p`), PLL multiplier (`pll_sys_m and pll_int_m`), and post scaler (`pll_sys_s and pll_int_s`) appropriately, you can get the necessary PLL output clocks (`PLL_OUT1 and PLL_OUT2`).

PLL can handle any external input clock in the range of 6 MHz to 64 MHz (27 MHz is default), and synthesize PLL_OUT1 between 40 MHz to 2500 MHz and PLL_OUT2 between 70.3125 MHz to 4000 MHz. For proper PLL operation, PLL input clock (`PLL_IN1 and PLL_IN2`) should be in the range of 6 MHz to 12 MHz. All PLL programming should be performed during software standby mode for stable system operation.

MIPI output is DDR using MIPI word clock, so that MIPI word clock is 1/16th of the bit clock. *Figure 6-2* illustrates the overall clock tree structure.

Interface PLL(PLL2) of S5K3B6SA is able to support SSCG(Spread Spectrum Clock Generation) to avoid EMI issues.

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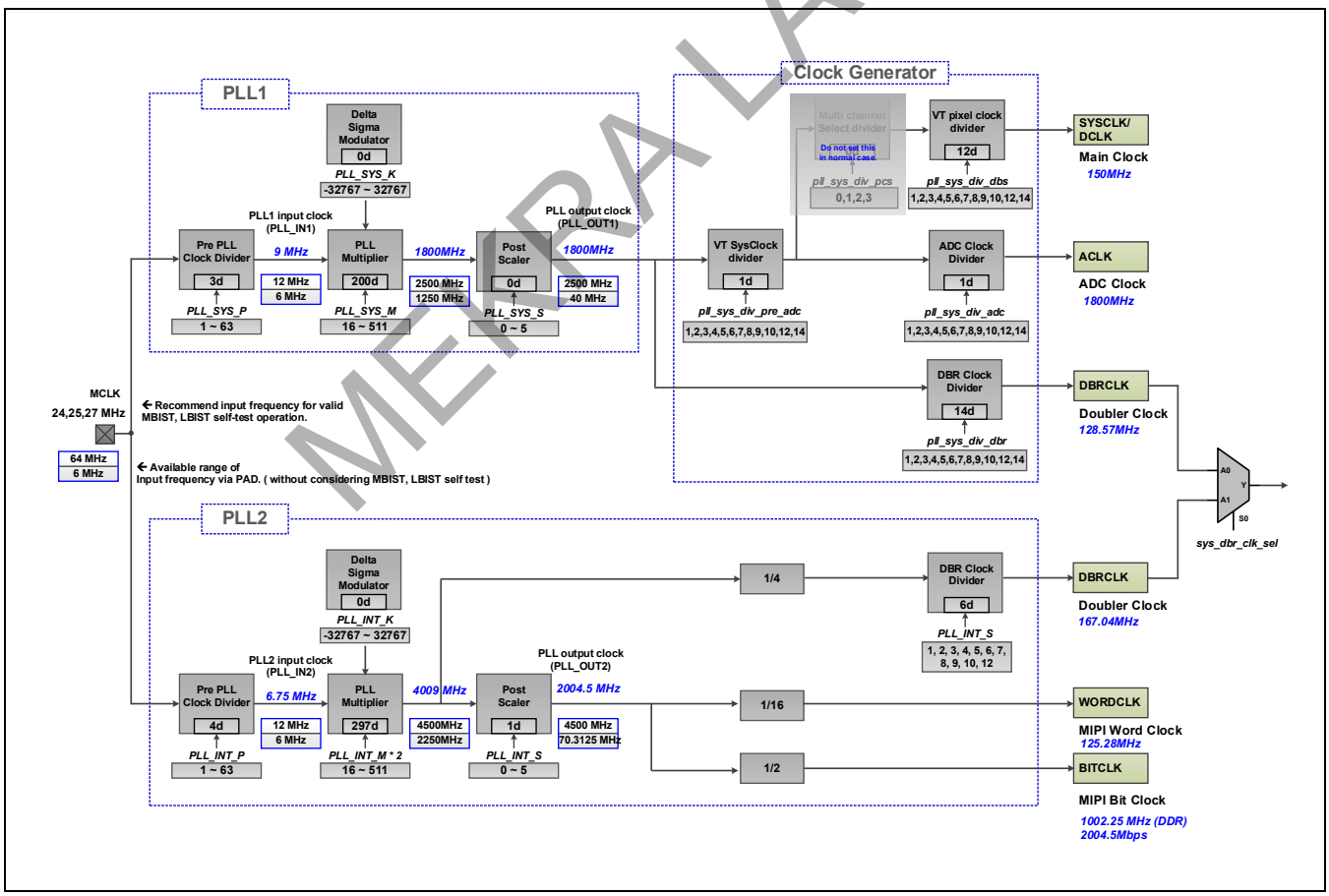


Figure 6-2 PLL and Clock Generator

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

NOTE:

All the register setting values are represented in decimal unit.

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Use the following equations to synthesize all necessary frequencies.

PLL1 Input Clock = External Input Clock (MCLK)/pll_sys_p (6 MHz to 12 MHz)
PLL2 Input Clock = External Input Clock (MCLK)/pll_int_p (6 MHz to 12 MHz)
PLL_OUT1 = (PLL_IN1×(pll_sys_m+pll_sys_k/65536))/2^pll_sys_s (40 MHz to 2500 MHz)
PLL_OUT2 = (PLL_IN2×(pll_int_m+pll_int_k/65536)*2)/2^pll_int_s
(70.3125MHz to 400 MHz, Bit clock divided by 2 for DDR)
1 ≤ pll_*_p ≤ 63 : 16 ≤ pll_*_m ≤ 511 : 0 ≤ pll_*_s ≤ 5 : -32767 ≤ pll_*_k ≤ 32767

6.3 Operation Mode

S5K3B6SA supports the operation modes described in *Table 6-1* along with their speed and output size.

Table 6-1 Typical Operation Modes

Mode	Interface	Output format	ADC [Bit]	Frame Rate [fps]	Bit Rate (Per Lane)	H-Pixel [ea]	V-Pixel [ea]	Comment
24b Combined		RAW12*2 *3)	12	60	1280	1936	1552	
20b Combined and Compress		RAW20	12	60	1120	1936	1552	
16b Combined and Compress		RAW16	12	60	960	1936	1552	Intra-scene DCG*2) and Lateral overflow with LFM (HDR4)
14 Combined and Compress		RAW14	12	60	800	1936	1552	
12 Combined and Compress		RAW12	12	60	640	1936	1552	
Line Re-order		RAW12	12	45	2080	1936	6208	
Pixel Re-order		RAW12	12	45	1920	7744	1552	

NOTE: 1. To improve the performance, the values of operation modes, output size, frame rate, and so on can be changed.

NOTE: 2. Sensor readout same pixel twice with difference conversion gain in frame.

NOTE: 3. The 24b Combined frame output in the RAW12 format where the horizontal pixel size doubled due to there is no defined output format of RAW24 in MIPI alliance.

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6.4 Slave Mode

To support multiple camera system, S5K3B6SA can operate in slave mode using the external sync. Vertical sync signal and the second CCI slave ID. Depending on the operation mode of the system, the direction of 'VSYNC_IN' PAD can be controlled as output or input mode.

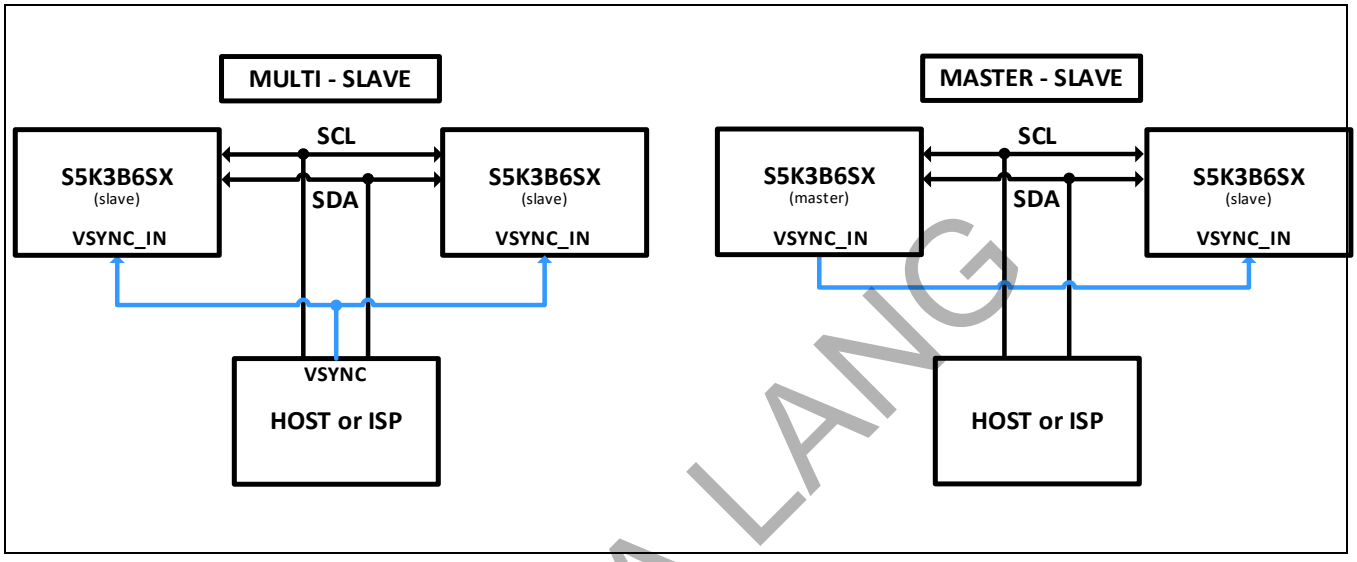


Figure 6-3 System Diagram for Multiple Camera

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6.5 Frame Readout Format

S5K3B6SA outputs up to 4 ea different (LPD HCG, LPD LCG, SPD CDS, and SPD LOF) frames through the line-interleaved WDR operation. During the WDR readout operation, the 1-row readout time of each exposed frame is tuned, and then the data of LPD HCG, LPD LCG, SPD CDS, and SPD LOF frame is out in every 4-row readout time period. This sensor can also operate in Intra-scene DCG with LFM mode.

6.5.1 Data Readout Frame Format

S5K3B6SA supports various data types and frame formats. *Figure 6-4* illustrates two kinds of output frame format. In line interleaving, LPD HCG, LPD LCG, SPD CDS, and SPD LOF row line data output in sequential order at every one row data output time. But in pixel reorder, LPD HCG, LPD LCG, SPD CDS, and SPD LOF data aligned and output in sequential pixel interleaving. In this case, whole pixel data in the same row is output over four times of the one row data output time in line interleaving.

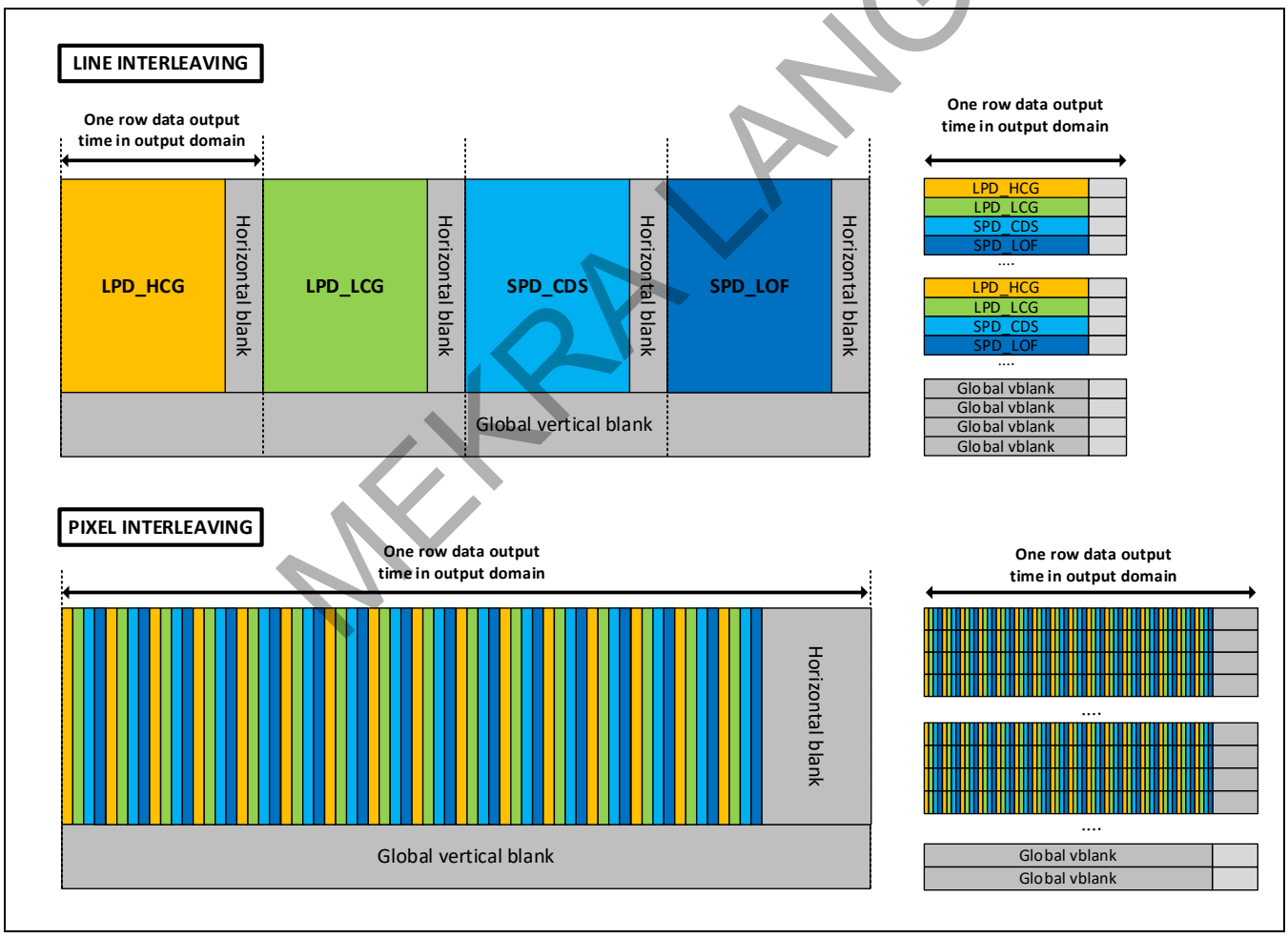


Figure 6-4 Line/Pixel Interleaved WDR Output Frame Format

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6.5.2 Data Combine

To reduce the amount of data transmission, S5K3B6SA can output combined 24-bit data instead of 4x12 bit raw data. Four 12-bit data are normalized to 24 bit by by using the internal algorithm.

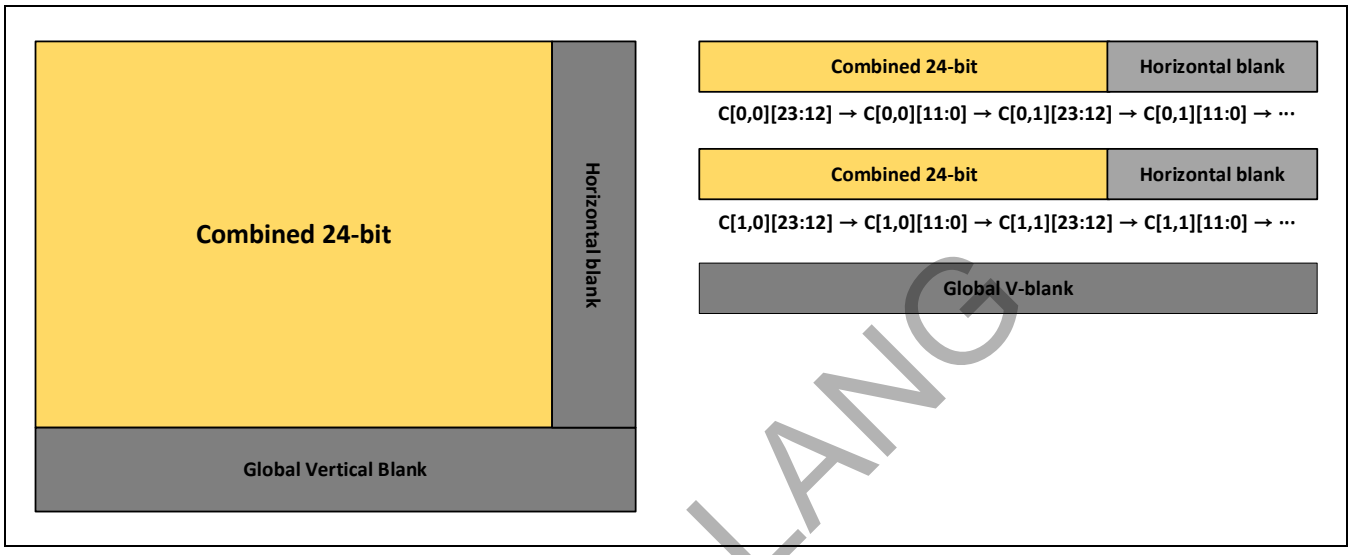
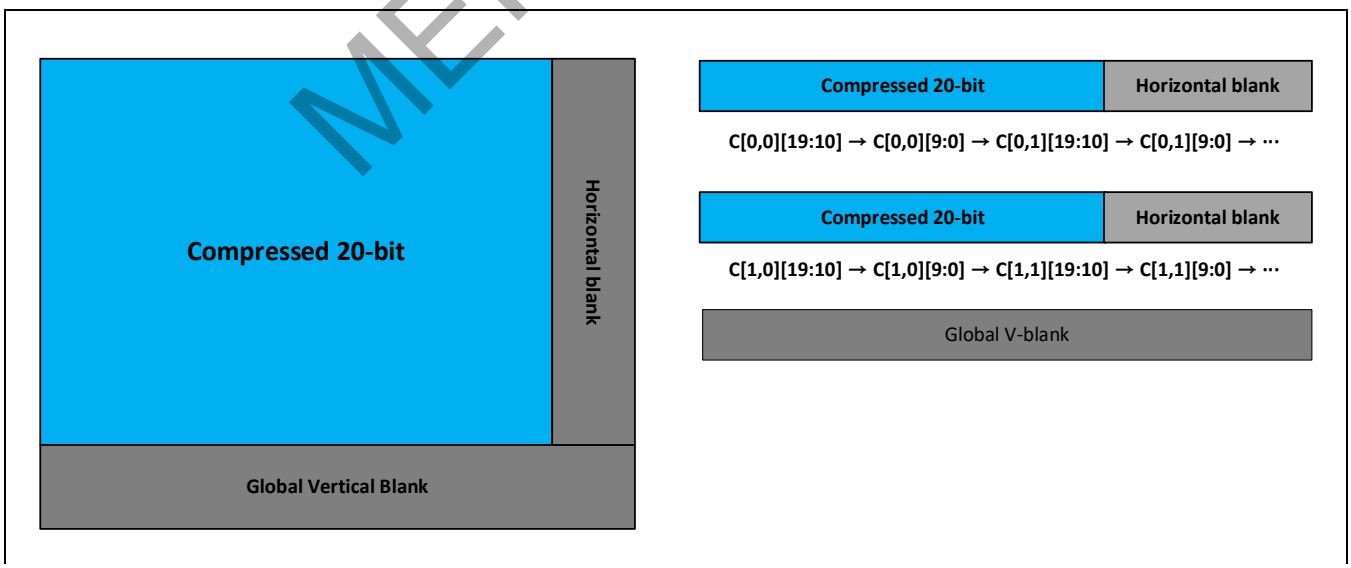


Figure 6-5 Combined Data Output Frame Format

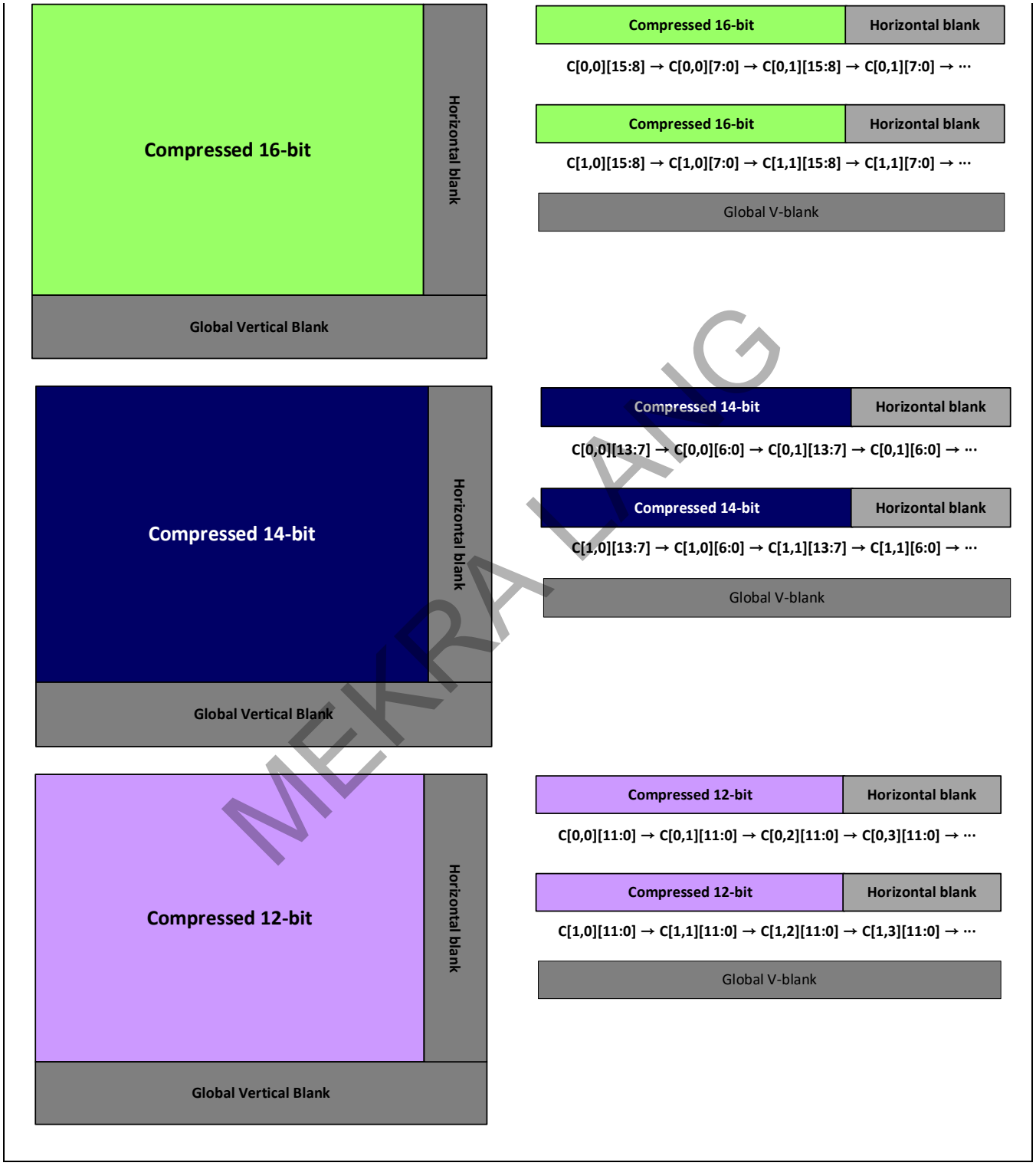
6.5.3 Data Compress

S5K3B6SA supports 20/16/14/12-bit compressed data output mode. The piecewise linear compression compresses the 24-bit combined data to 20/16/14/12-bit data with 32 knee-points. These knee-points are controlled by registers.



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Figure 6-6 20/16/14/12-Bit Compressed Data Output Format

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6.6 Image Orientation (H/V)

The pixel data are normally readout from left to right in horizontal direction and from top to bottom in vertical direction. By changing the mirror/flip mode, the readout order is reversed and the resulting image is flipped as a mirror image. Pixel data outs from right to left in horizontal mirror mode and from bottom to top in vertical flip mode. The horizontal mirror and the vertical flip modes are programmed by the image orientation register.

Following are the four pixel readout orders supported by sensor:

- **Standard readout:** Horizontal start address and width registers control the addressed region of horizontal pixel data, and vertical start address and width registers control the addressed region of vertical pixel data.
- **Horizontally mirrored readout:** Horizontal start address and orientation registers control the addressed region of horizontal pixel data output, and the addressed region of vertical pixel data output is the same as that of the standard readout.
- **Vertical flipped readout:** The addressed region of horizontal pixel data output is same as that of the standard readout, and vertical start address and orientation registers control the addressed region of the vertical pixel data output.
- **Horizontally mirrored and vertically flipped readout:** Horizontal start address and orientation registers control the addressed region of the horizontal pixel data, and vertical start address and orientation registers control the addressed region of the vertical pixel data.

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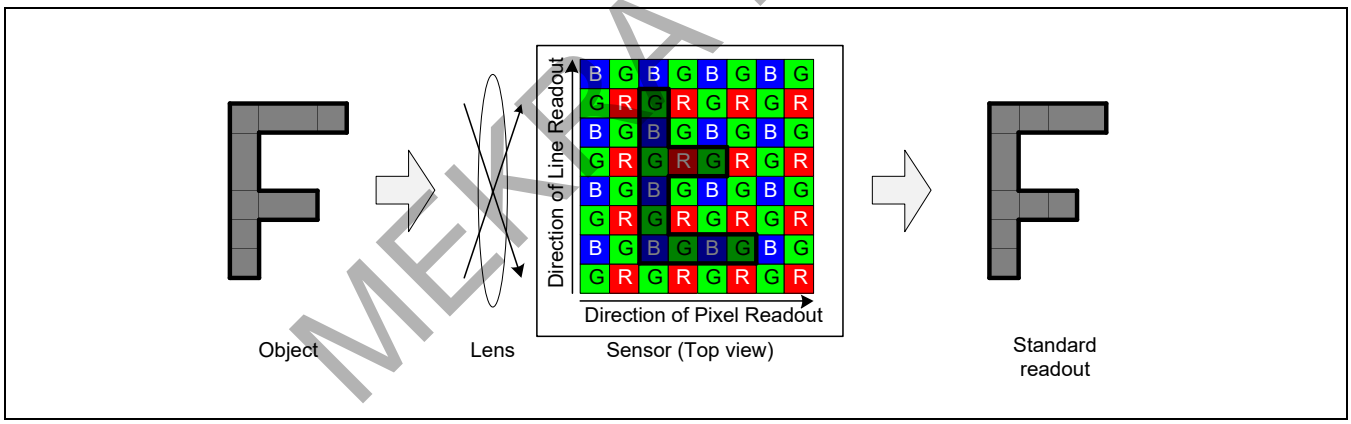
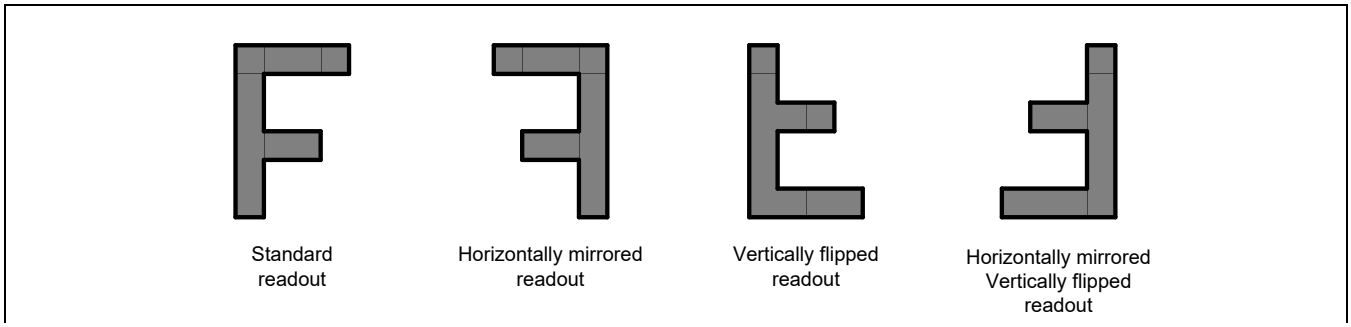


Figure 6-7 Object, Sensor and displayed image in Standard Readout



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Figure 6-8 Four readout orders and displayed images

6.7 Frame Rate Control

The line rate and frame rate are changed by varying the size of the virtual frame. The number of clocks for horizontal readout and number of lines for vertical readout that consist of the virtual frame can be controlled by registers. The horizontal and vertical blanking times (horizontal blanking time: # of clocks for horizontal readout columns and vertical blanking time: # of lines for vertical readout lines) should be larger than minimum vertical blank time of sensor operation.

Use the following equation to calculate the frame rate.

Frame rate = SYSCLK / (# of clocks for horizontal readout[1)] * # of lines for vertical readout[2)])

[1)] # of clocks for horizontal readout = line_length_pck
[2)] # of lines for vertical readout = frame_length_lines ("frame length lines" value should divide into 3 at intra-scene DCG with LFM mode and 4 in the intra-scene DCG with LFM mode.)

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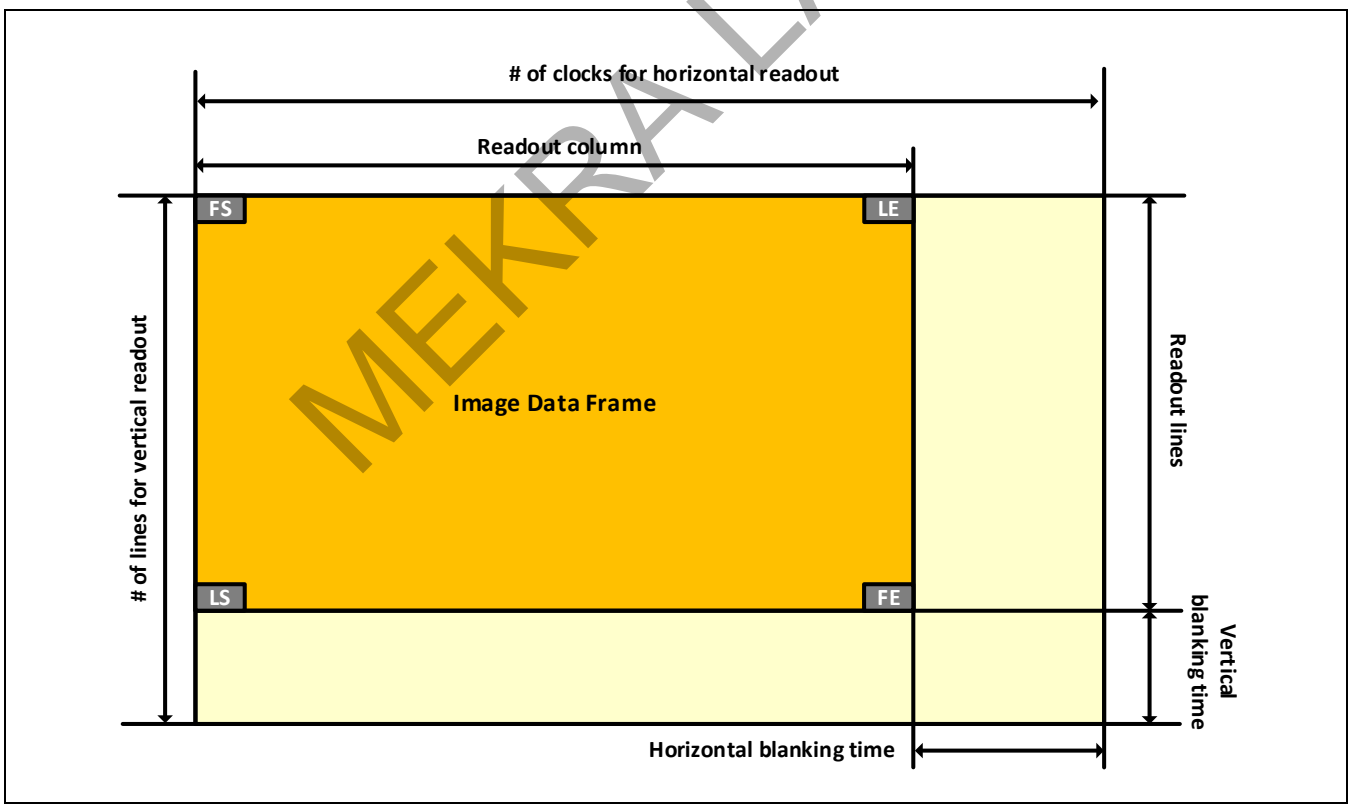


Figure 6-9 Example Format of Virtual Frame

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6.8 Integration Time Control (Electronic Shutter Control)

The pixel integration time is controlled by the shutter operation. During the shutter operation, the amount of integration time is determined by the column step integration time control register (fine_integration_time) and the line step integration time control register (coarse_integration_time). Use the following formula to calculate the total integration time of the sensor.

```
Total_integration_time = {coarse_integration_time x # of clocks for horizontal readout[1)]+
fine_integration_time}/(SYSCLK) [sec]

[1)] # of clocks for horizontal readout = line_length_pck
```

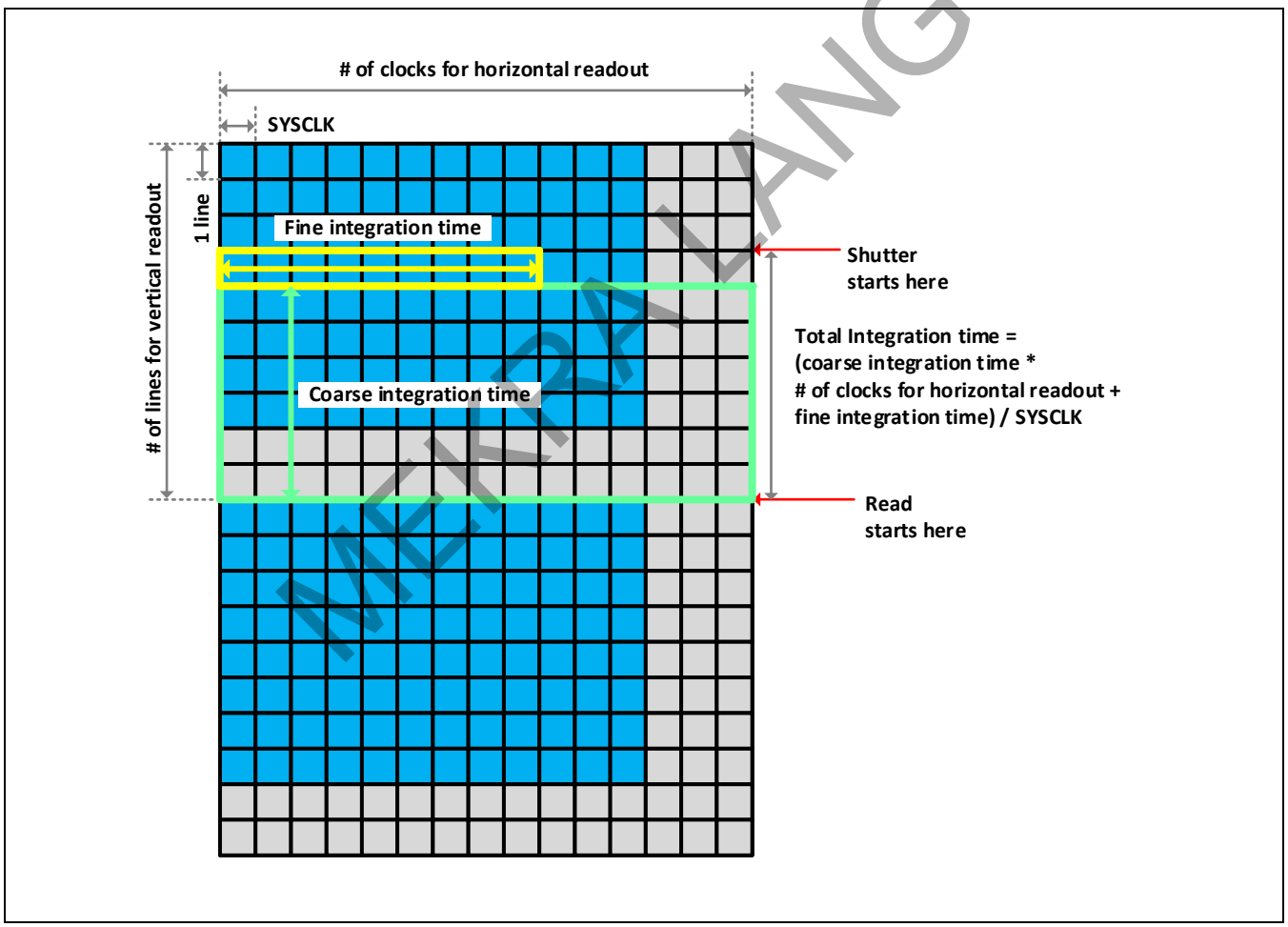


Figure 6-10 Example of Calculating Integration Time

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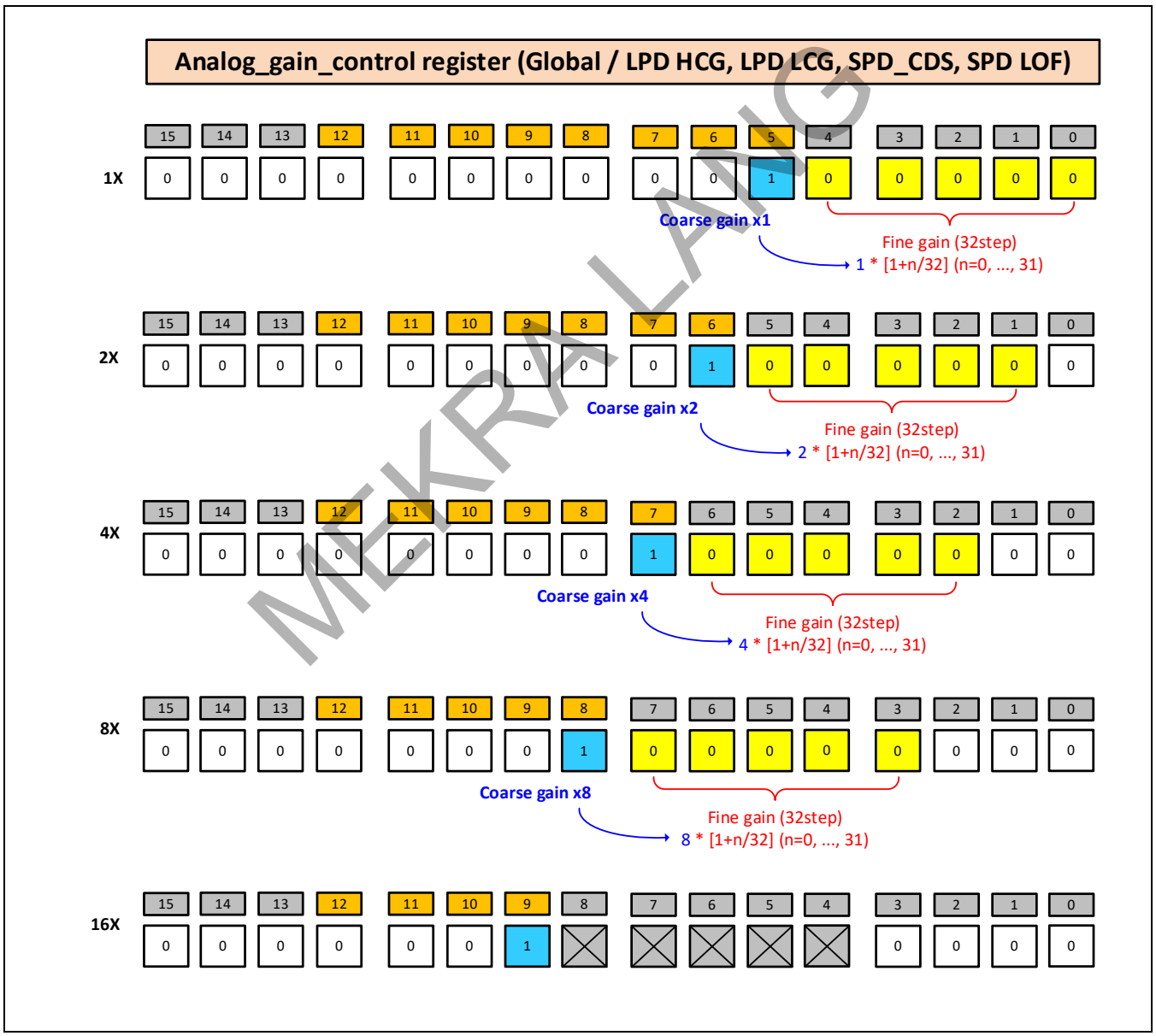
Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.9 Gain Control

6.9.1 Analog Gain

User can control the gain of pixel signal by analog gain control register(`analogue_gain_code_global`). User can only control the analog gain of LPD HCG frame, and use the following equation to control the analog gain.

Analog Gain
Coarse step x2, **Fine step: x1/32 (@ coarse gain 2^n ~ 2^n+1, n = 0, 1, 2, 3)**



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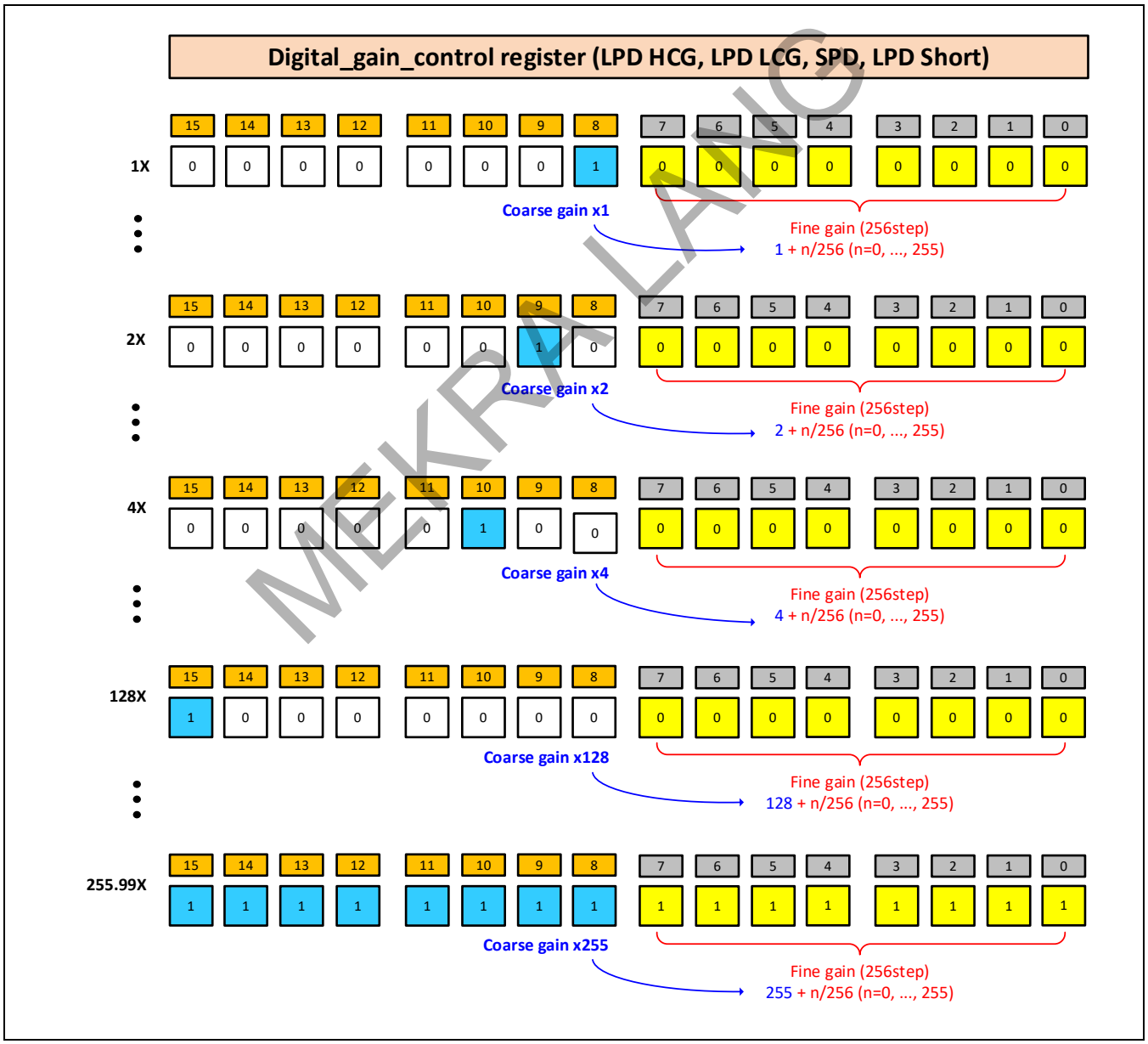
Figure 6-11 Example of Analog Gain Control

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.9.2 Digital Gain

You can control the gain of pixel signal for each channel or each exposure type by digital gain control register. As a result, use the following equation to control the digital gain.

- Each exposure type
Digital gain = exposure_gain[15:0]/256
- Each channel
Digital gain = channel_gain[11:0]/4096



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Figure 6-12 Example of Digital Gain Control (Exposure Type)

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6.10 Auto Dark Level Correction

Auto Dark Level Correction (ADLC) is responsible for row noise reduction and frame black level recovery by subtracting a weighted sum of optical black pixels.

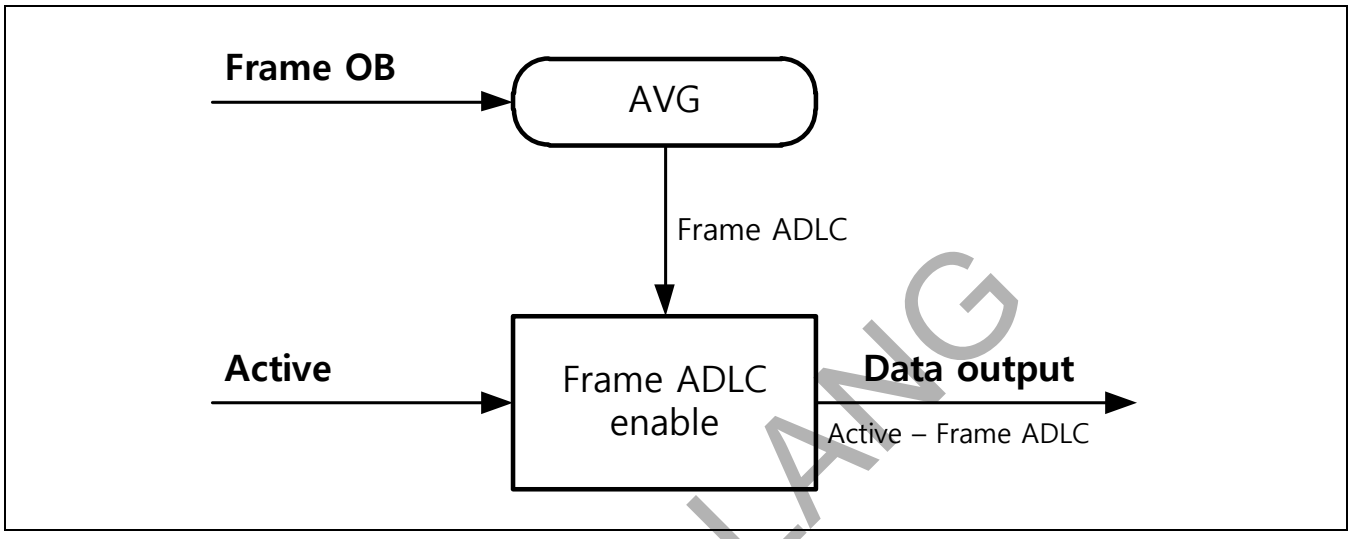


Figure 6-13 ADLC Algorithm

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6.11 Image Output Format

6.11.1 MIPI

S5K3B6SA MIPI CSI-2 interface is a four-lane high-speed serial interface to connect the camera sensor to a host processor. S5K3B6SA MIPI core IP is compatible with the MIPI alliance standards for D-PHY and CSI2 v2.1.

The maximum high-speed clock frequency of the MIPI is 4.0 GHz.

MIP supports following features:

- MIPI DPHY CSI-2: 4-lane (2/3/4-lane modes) with maximum of 4.0 Gbps (bit clock rate)
- RAW8, RAW10, RAW12, RAW14, RAW16, RAW20 data types and RAW24 data types
- MIPI transmitting (Tx) only
- One clock lane
- Upto eight number of virtual channels for the multiple exposure mode
- Ultra Low Power State Mode (ULPS)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

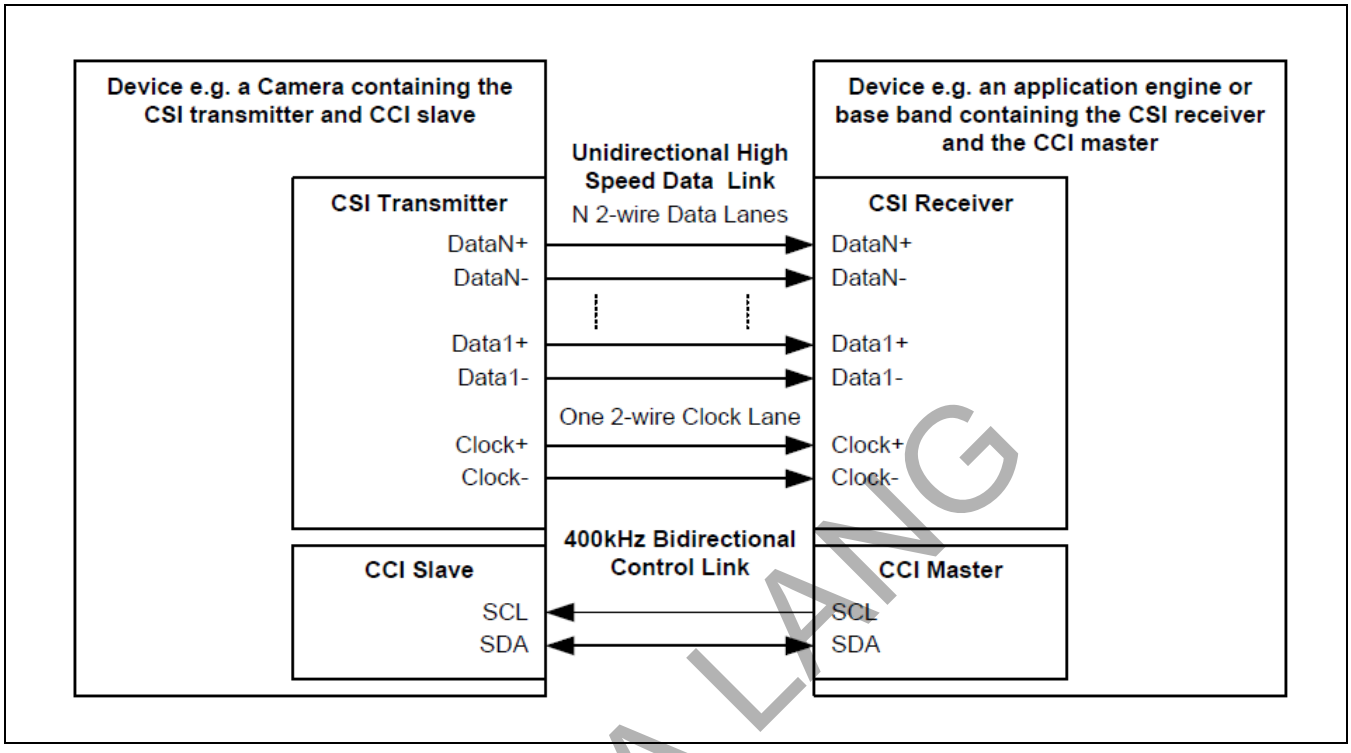


Figure 6-14 CSI-2 and CCI Transmitter and Receiver Interface (Reference from MIPI CSI-2 v2.1)

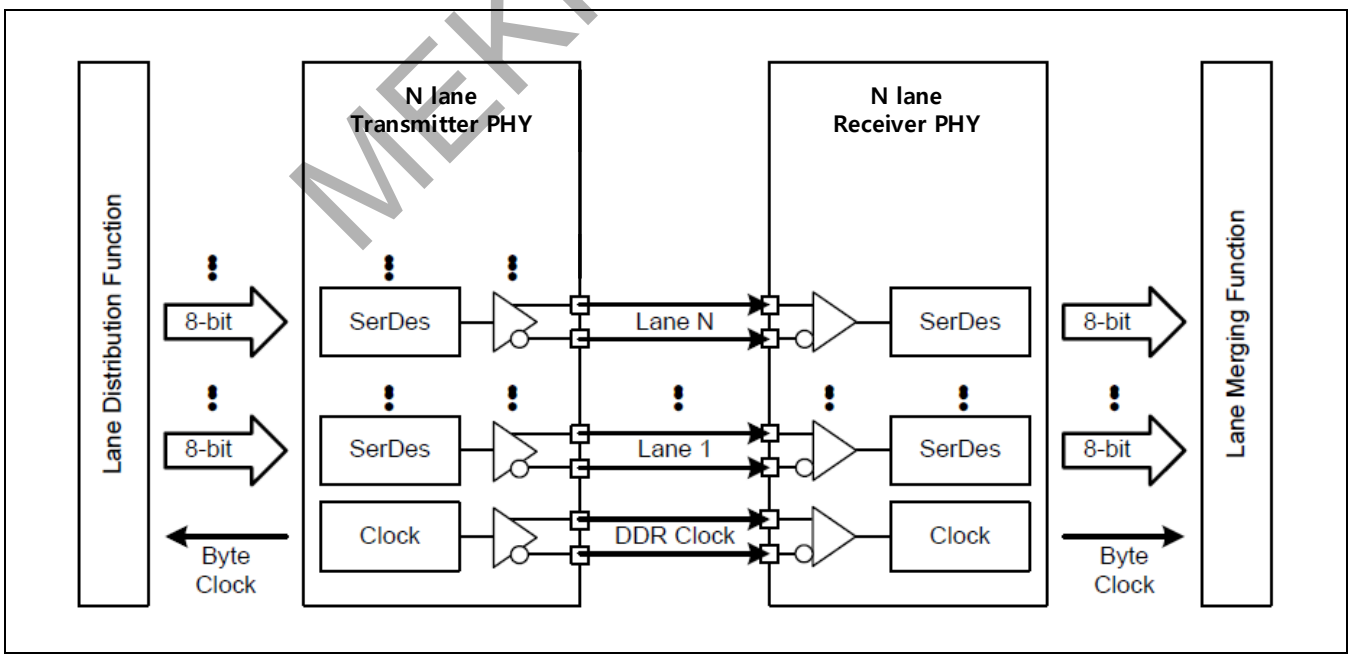


Figure 6-15 Example of N-Lane Transmitter and N-Lane Receiver (Reference from MIPI CSI-2 v2.1)

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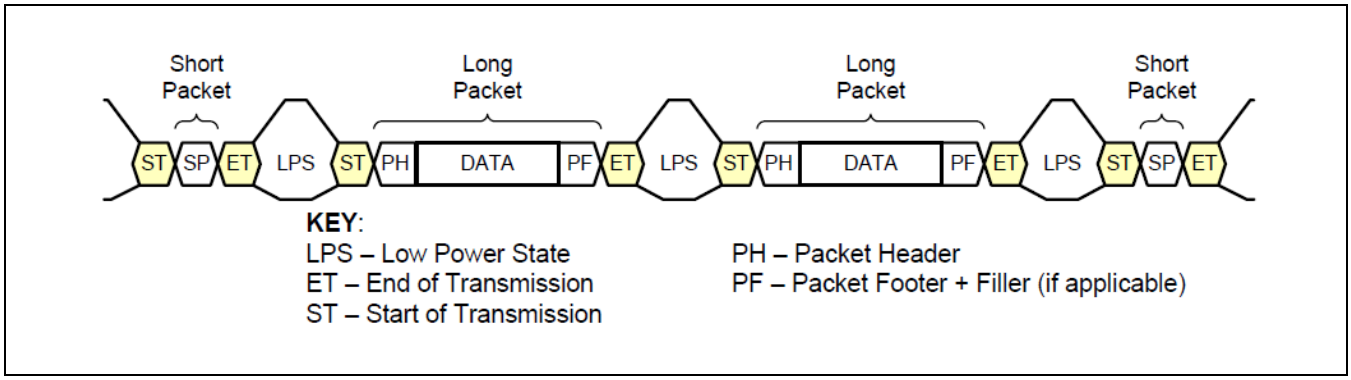


Figure 6-16 Low-Level Protocol Packet Overview (Reference from MIPI CSI-2 v2.1)

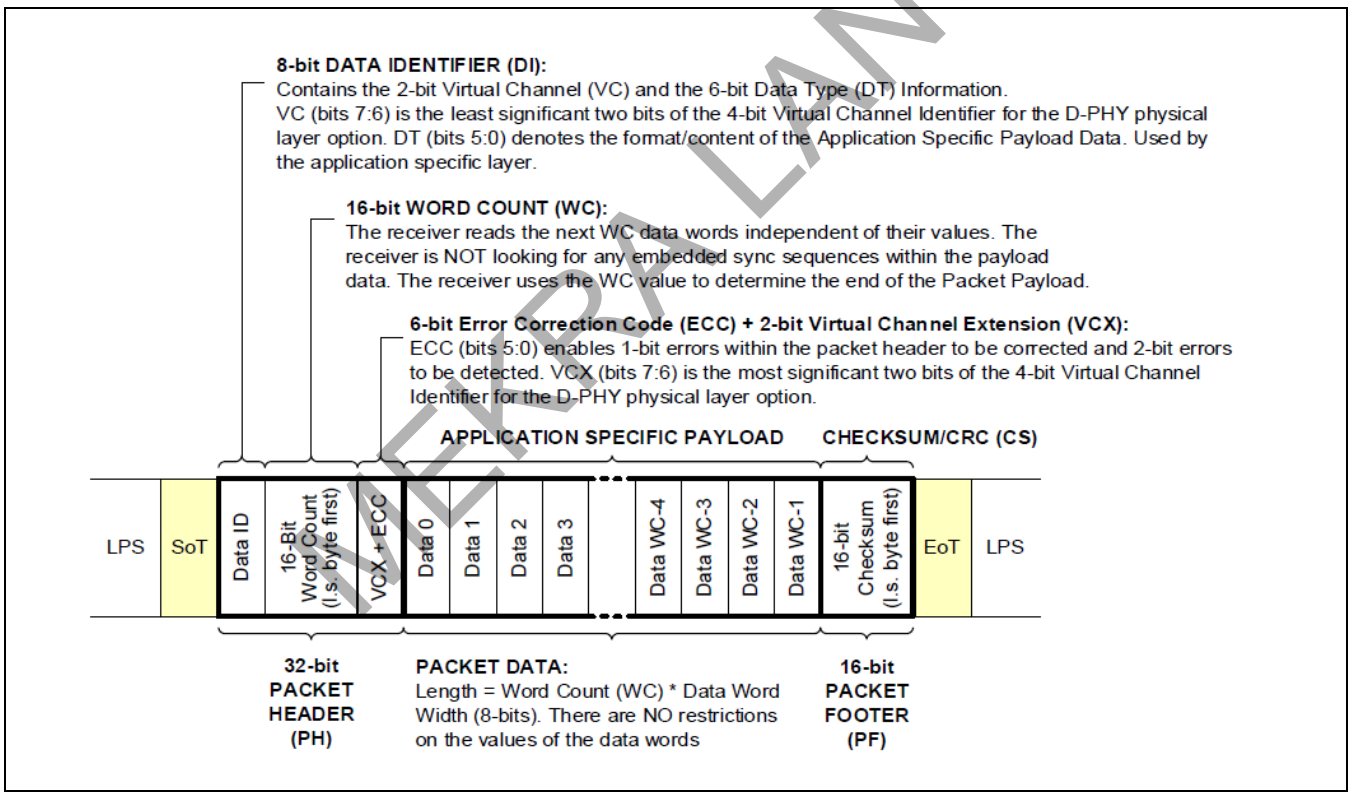


Figure 6-17 Long Packet Structure (Reference from MIPI CSI-2 v2.1)

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Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

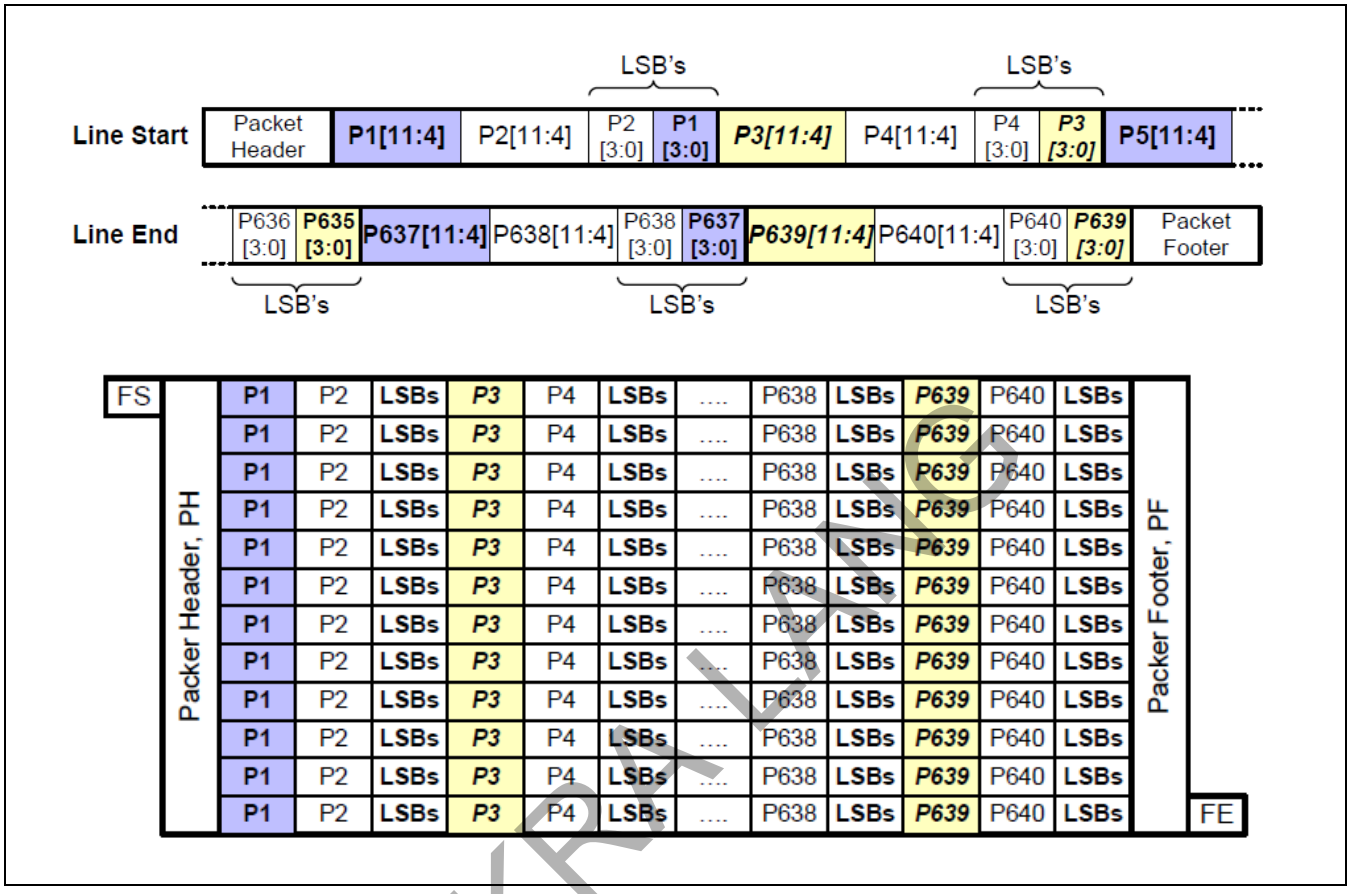


Figure 6-18 RAW12 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

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Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

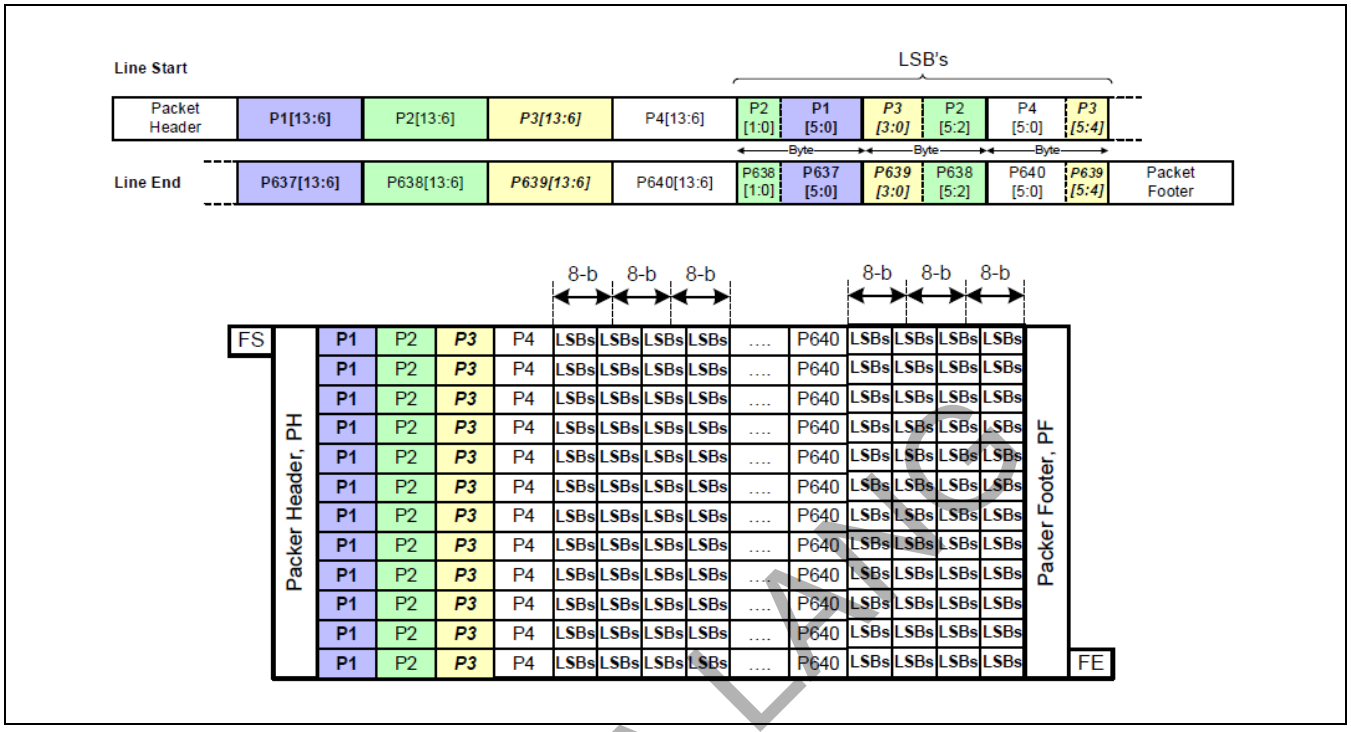
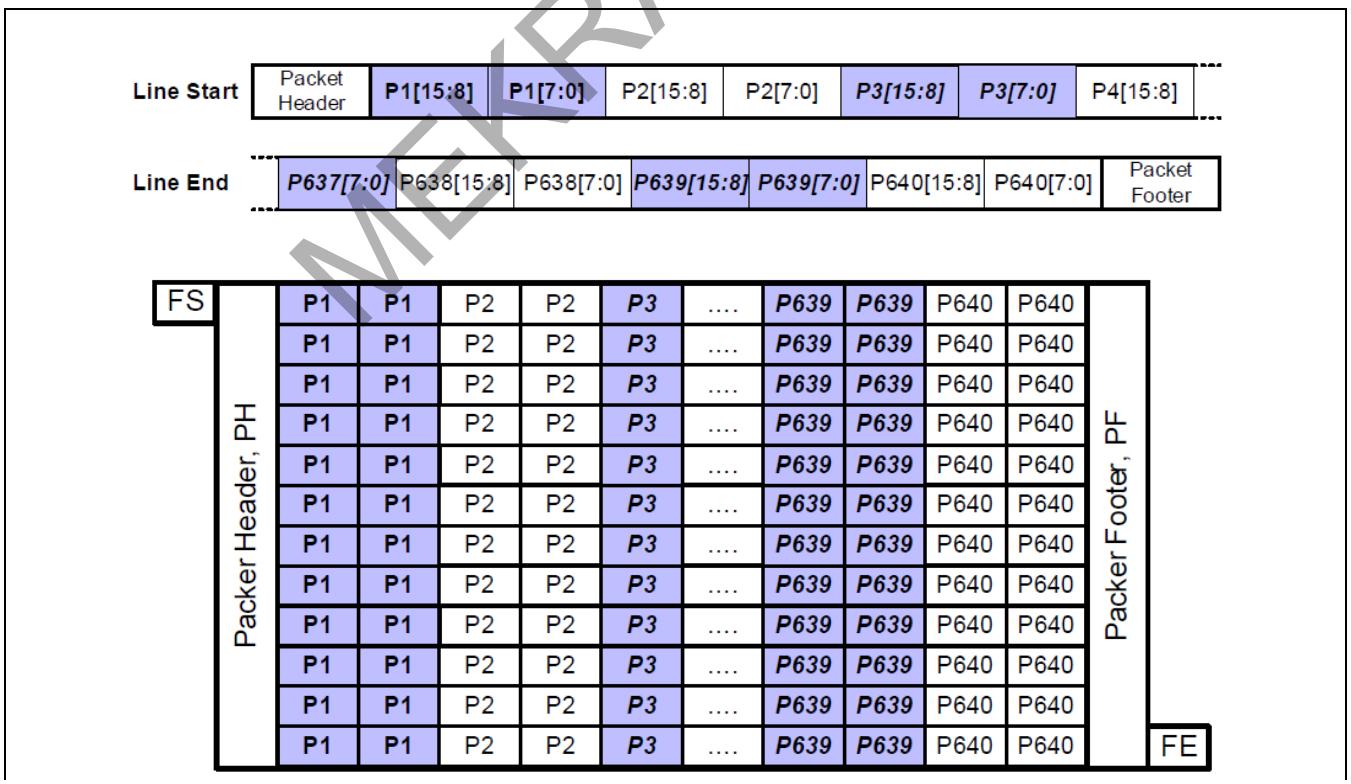


Figure 6-19 RAW14 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)



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Figure 6-20 RAW16 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

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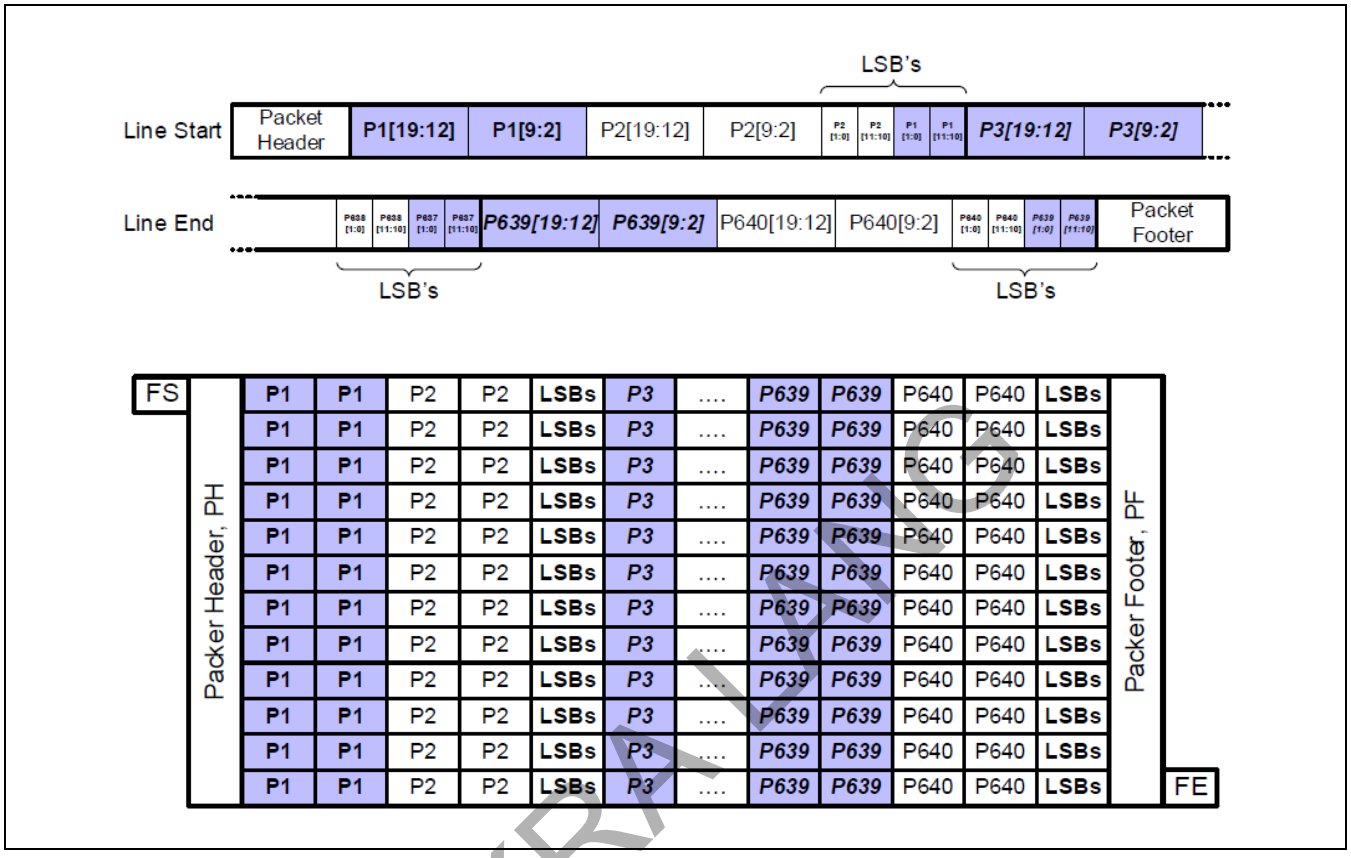


Figure 6-21 RAW20 Transmission and Frame Format (Reference from MIPI CSI-2 v2.1)

MEKRA LABS

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.1.1 MIPI Virtual Channel Structure

S5K3B6SA outputs various types of data such as embedded header , embedded footer, combined and compressed main data, RAW and defect and detection map for the reference of back-end processor. Therefore, S5K3B6SA can apply virtual channel to the various types of data.

Table 6-2 describes the various types of data which can output within the same frame and apply the virtual channel at each data when sensor operates in intra-scene DCG with LFM mode.

Table 6-2 Examples of Applying Virtual channel at each data at intra-scene DCG with LFM mode.

Data Type	Output Mode	Data1		Data2		Data3		Data4	
		Data type	VCI	Data type	VCI	Data type	VCI	Data type	VCI
Embedded Header		0x12	O	-					
Main Frame	Line interleaving	RAW12	O	RAW12	O	RAW12	O	RAW12	O
	Pixel interleaving	RAW12	X	-					
	Partially Combined	RAW20	O	-		RAW12	O	Map (UD: 0x32)	O
		RAW16	O	-		RAW12	O	Map (UD: 0x32)	O
	Combined	RAW24	O	-				Map (UD: 0x32)	O
	Compress	RAW20	O	-				Map (UD: 0x32)	O
		RAW16	O	-				Map (UD: 0x32)	O
		RAW14	O	-				Map (UD: 0x32)	O
		RAW12	O	-				Map (UD: 0x32)	O
Embedded		0x30	O	-					

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Footer				
Statistics		0x31	O	-

O : Supported / X : Not Supported / UD : User-Defined

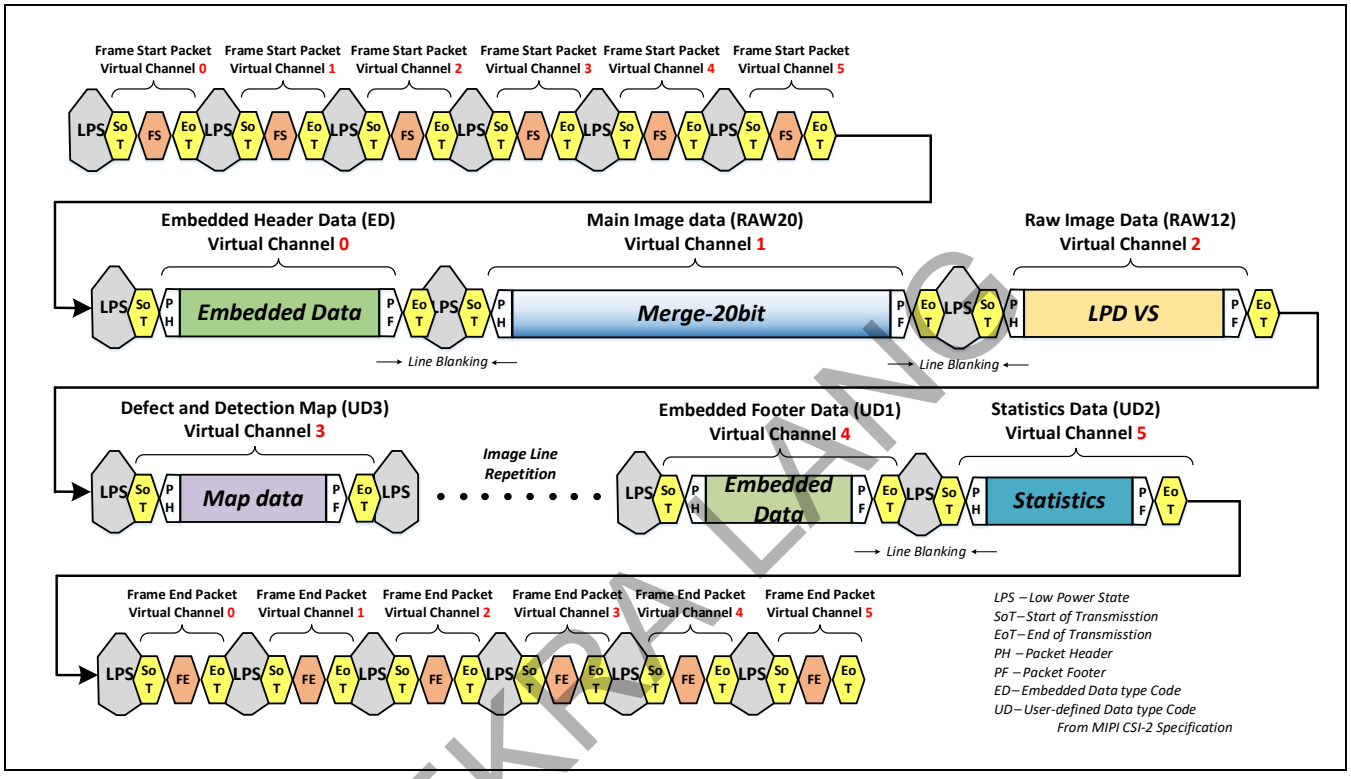


Figure 6-22 Example of MIPI Interleaved Virtual Channel and Data Type

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6.11.2 ULPS Mode

S5K3B6SA MIPI Lanes will enter ULPS mode automatically at every software standby mode. DPHY can reduce power consumption by entering ULPS mode.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

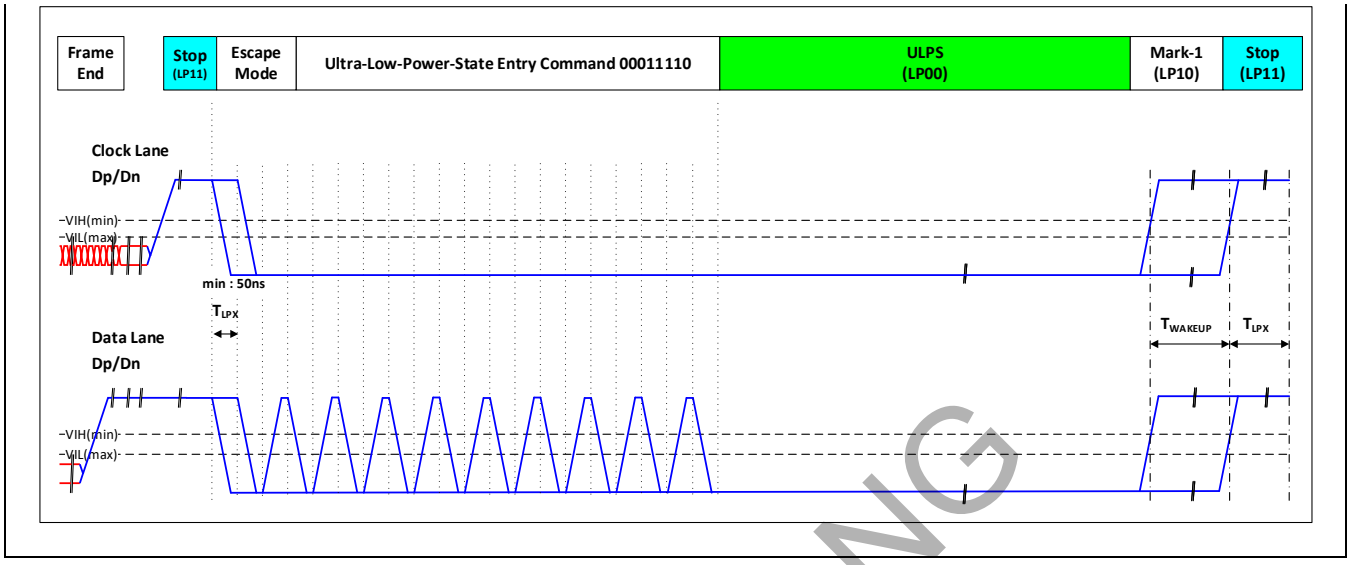


Figure 6-23 DPHY Ultra-Low-Power State Sequence

6.12 Image Compensation functions

S5K3B6SA provides some features to support image output with better characteristics.

6.12.1 Bad Pixel Correction

S5K3B6SA supports 'Static and Dynamic Bad Pixel Correction' feature. There needed to write Coordinate and Pattern information of Static BP to One Time Programmable memory and have capability of cure BP up to 1000 number at Large Pixel and 700 number of BP at Corner Pixel.

For Dynamic Bad Pixel Correction, S5K3B6SA support individual control via register such as enable/disable and threshold values which define intensity of correction at each exposure frame up to 4.

6.12.2 Lens Shading Correction

S5K3B6SA support Lens Shading Correction feature to reduce differences between shading curves of edge of image and center by applying a different gain on each exposure frame up to 4.

6.12.3 Noise Reduction

S5K3B6SA supports noise reduction for SPD and Merged frame. the denoising concept which is based on the pattern matching technique.It based on large area scan and search for pattern similar to that pattern in the central pixel. All the central pixels of the similar patterns are averaged producing the new pixel value.

6.13 Miscellaneous functions

6.13.1 Crop

S5K3B6SA allows cropping an image and padding at the end with a predefined value before or after Merger.

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6.13.2 OTP

One-Time Programmable memory is embedded in S5K3B6SA. OTP consists of 256 pages (16384 bytes) and every single page has 32 data buffers. User can access reserved pages (4864 bytes) for customer usage by I2C protocol.

6.13.3 Temperature Monitor

S5K3B6SA supports temperature monitor which provides junction temperature data in imager to host. There are 2 temperature monitor are implemented in imager. The temperature monitor is performing calibration which is possible inside the imager, and an error flag can also be sent if the temperature is out of the set range. In order to get valid value of temperature data of image sensor, values which needed at calibration should be written in OTP at manufacturing stage.

6.13.4 Voltage Monitor

S5K3B6SA supports voltage monitors which supply voltages and internally generated voltages. voltage monitor converts a voltage to a digital code, with which the comparison logic checks whether the input voltage is out of range. If the input voltage is out of range, an error flag gets active. There are two voltage monitors in the imager. Each voltage monitor generates the reference voltage independently from each other. In order to get valid value of voltages which operate image sensor, values which needed at calibration should be written in OTP at manufacturing stage.

6.13.5 LBIST

S5K3B6SA supports LBIST(Logic Built-in self-test) which is scan stuck-at test for digital logic cirtuit every boot-up sequence (when External Reset released) automatically.

6.13.6 MBIST

S5K3B6SA supports MBIST(Memory Built-in self-test) which is memory test for internal SRAM every boot-up sequence (when External Reset released) automatically.

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7 Embedded Data and Safety Features

This chapter includes the following sections:

- Embedded data (Header and Footer)
- Safety features for ASIL-B systems

7.1 Embedded Data

There are several rows of data prepended (embedded header) and appended (embedded footer) to each frame of image data for safety.

The embedded data consist of based on CSI-2 (RAW8 packed data) specifications. The packing format for embedded data can be same as image format optionally.

Table 7-1 List of Output Data

Items	Contents
Embedded Header	
Frame information	Horizontal and vertical size, Frame counter
Image size	Horizontal and vertical start and number of readout columns and lines.
Image orientation	Mirror and flip
Sub-sampling ratio	Horizontal and vertical even and odd increments
Register value for AE	Integration time, analog gain, and digital gain
Register value for WDR	Number of each 32 X and Y knee point registers. Registers which relate with WDR algorithm.
Optical black pixel data	Average of optical black pixels
Data format information	RAW 8/RAW10/RAW12/RAW14/RAW16/RAW20
Embedded Footer	
ASIL	Result of internal safety features.
Security	Signature value of frame
Statistics	Histogram

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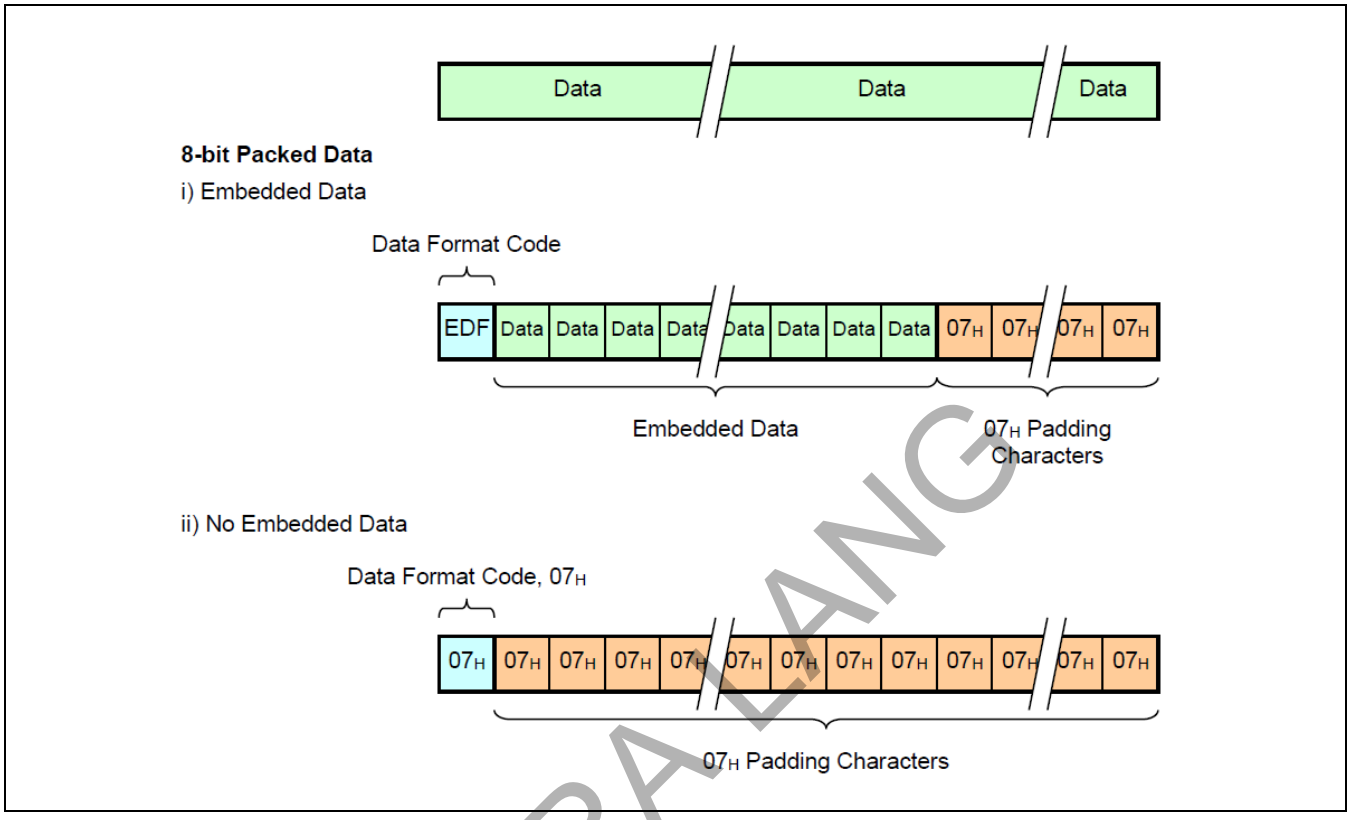


Figure 7-1 Data Output Stream (Reference from GIIC v0.8)

7.2 Safety Features for ASIL-B Systems

S5K3B6SA provides safety features for vision systems to meet ASIL-B grade.

Table 7-2 Features for ASIL-B Systems

Features	Description	Fault Detection Period/Time
Memory fault (BIST)	SRAM read-back test detects the stuck at 1 or 0 faults.	EDS (Wafer test)
Memory fault (POST)	SRAM read-back test detects the stuck at 1 or 0 faults when every RSTN releases from external.	At every boot-up sequence
Memory ECC	All SRAMs include Error Correction Code (ECC) for 1-bit error correction and 2-bit error detection.	At every frame
Control interface fault	CRC value of CCI data (from host) are put to the embedded data lines. (CRC value is available for enabling CCI CRC.)	User-defined
Output interface fault	Image data transmission to ISP or AP contains CRC	At every row

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Features	Description	Fault Detection Period/Time
	of data to detect errors in communication.	
Clock monitoring	Monitoring frequency of clock from system PLL and interface PLL after the phase locked.	User-defined
E2E protection	End-to-end protection of image data and meta data (embedded data)	At every frame
Embedded data (header/footer)	There are several rows of data which are prepended to the image frame output. The image frame output consists of registers and ASIL supplement data.	At every frame
Digital frame counter	This value is included as part of the embedded data (64 bits) to track the frame count.	At every frame
Test patterns (digital)	Digital test pattern is prepended or appended to each frame of image data.	At every frame/On-demand
Cyber security	Sensor authentication at boot-up sequence and secured CRC during data streaming.	At every frame
Voltage monitoring	Detects the abnormal operation of internal voltage such as stuck to high or low.	During data streaming
Temperature sensor	Temperature sensor reports the temperature in degrees celsius from -40°C to 125°C within the embedded line data.	At every frame
Pixel control line	Checks the control signal of pixel transistors.	At every frame
ADC	Checks the ADC operation of all columns.	At every frame

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8 Electrical Characteristics

This chapter includes the following sections:

- Absolute maximum rating
- Operating conditions
- DC characteristics
- AC characteristics

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8.1 Absolute Maximum Rating

Table 8-1 Absolute Maximum Rating (TBD)

Description	Symbol	Min.	Typ.	Max.	Unit
Digital absolute Max.	VDDD (Max.)	–0.5	–	1.5	V
Analog absolute Max.	VDDA (Max.)	–0.3	–	3.6	
I/O absolute Max.	VDDIO (Max.)	–0.3	–	2.5	
Digital input voltages absolute Max (1)	VIP (Max.)	–0.3	–	VDDIO + 0.3	
VCAP analog voltage absolute Max (2)	VCAP	–0.3	–	4.2	
Storage temperature	TSTR	–40	–	-40 to +125	°C

NOTE:

The values beyond the absolute maximum ratings damage the devices. They do not imply any functional operation of the device at these conditions or any other conditions beyond the recommended operating conditions.

1. Digital Inputs: MCLK, RSTN, SCE, SDA, SCL, GPIO_*, VSYNC_IN, HSYNC_IN
2. Voltage on external analog capacitors

8.2 Operating Conditions

Table 8-2 Operating Conditions

Description	Symbol	Min.	Typ.	Max.	Unit
Analog supply voltage	VDDA	2.66	2.8	2.94	V
I/O supply (low voltage)	VDDIO	1.7	1.8	1.9	V
Digital supply	VDDD	0.95	1.05	1.15	V

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Operating temperature	TOPR (1)	-40	-	125	°C

NOTE:

1. TOPR is the junction operating temperature of sensor function.

8.3 DC Characteristics

Table 8-3 DC Characteristics

Description	Symbol	Min.	Typ.	Max.	Unit
HIGH level input voltage (LVCMOS interface) VDDIO = 1.8 V	VIH	0.7 × VDDIO*	-	-	V
Low level input voltage (LVCMOS interface) VDDIO = 1.8 V	VIL	-	-	0.3 × VDDIO	V
Output high voltage Test Condition(IOH =1.5 mA, 3 mA, 6 mA, and 9mA)	VOH	0.8 × VDDIO	-	VDDIO	V
Output low voltage Test Condition(IOL =1.5 mA, 3 mA, 6 mA, and 9mA)	VOL	VSSIO	-	0.2 × VDDIO	V
Input leakage current (1)	IIH (Vin = VDDIO) (2)	-	-	15	μA
	IIL (Vin = VSSIO) (2)	-15	-	-	μA
Input capacitance	CIN (3)	-	-	5	pF
Input operating frequency	FIN (MCLK)	6	27	64	MHz
Analog supply operating current (VPIX, VDDA, and VDDA_MIPI)	IDDA	-	50.35(4)	62(5)	mA
Digital supply operating current	IDDD	-	164.45(4)	337(5)	mA
I/O supply operating current	IDDIO	-	10.99(4)	12.7(5)	mA

NOTE:

1. When the sensor package is soldered on PCB, IIH, and IIL of SDA, SCL pins depend on the value of the off-chip pull-up devices such as resistors. This values are used for low I/O voltage scenarios (VDDIO=1.8 V Typ. voltage).
2. Vin means the input voltage of each package pin.
3. CIN max. can be varied based on the environment.
4. Condition of Typ. operating current: At Typ. voltage, Tj 85 ℃, 30fps, 325 Mbps MIPI 4-lane
5. Condition of Max. operating current: At Typ. voltage, Tj 125 ℃, 30fps, 325 Mbps MIPI 4-lane
   Total power is managed below 550mW.

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8.4 AC Characteristics

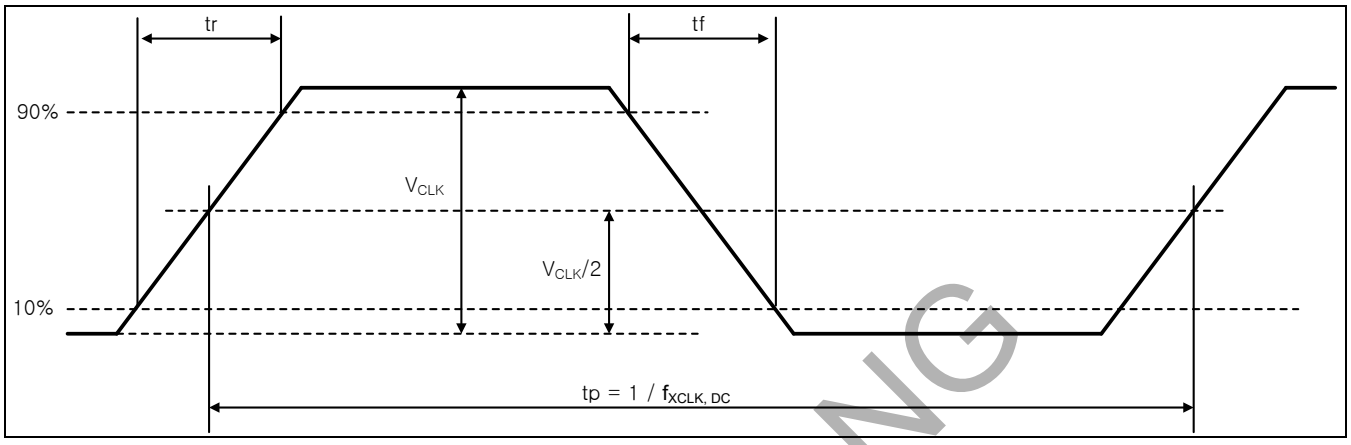


Figure 8-1 Master Clock Waveform

Table 8-4 AC Characteristics

Characteristics	Symbol	Condition	Min.	Typ.	Max.	Unit
External clock frequency (NOTE)	$f_{XCLK, DC}$	-	6	-	64	MHz
External clock duty cycle (NOTE)	f_{XDUTY}	-	40	-	60	%
External clock period jitter (peak-to-peak)	T_{jitter}	-	-	-	600	ps
External clock rise and fall times	-	-	-	-	6	ns
PLL locking time	t_{LOCK}	-	-	-	500	µs

NOTE: It is applicable for MCLK pin.

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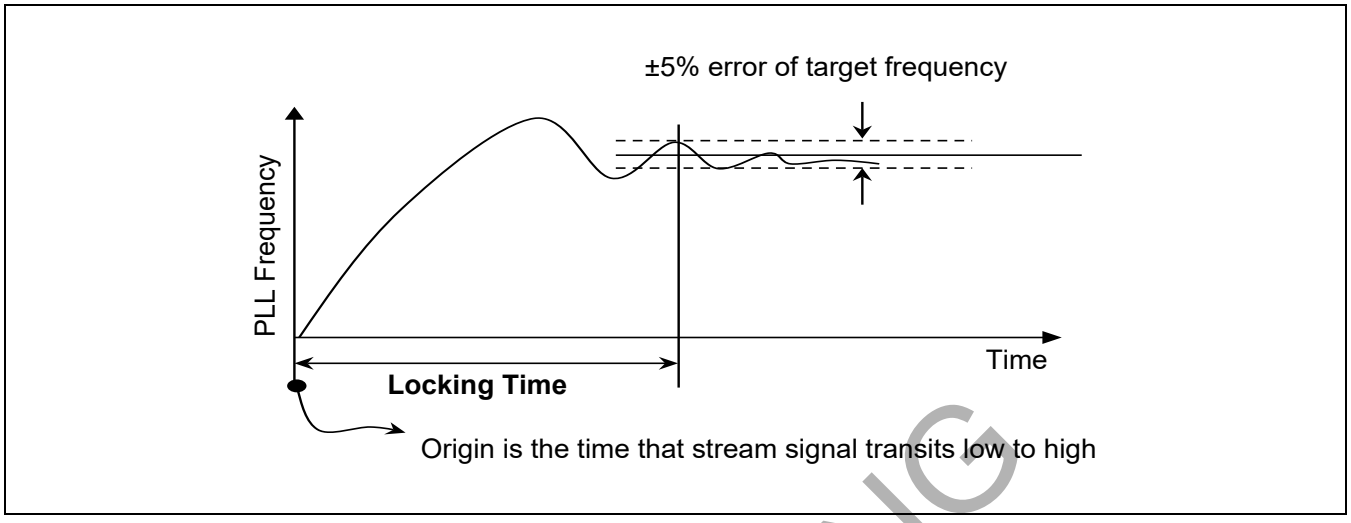


Figure 8-2 PLL Locking Time

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